

COMPLETE CLASSIFICATION OF REFLECTION MATRICES FOR SIX DIFFERENCE-FORM SPIN-CHAIN R -MATRICES

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ABSTRACT. An integrable quantum spin chain acquires an integrable open end when a reflection matrix solves the boundary Yang–Baxter equation of the chain’s bulk R -matrix. For the six regular difference-form 4×4 R -matrices that fall outside the eight-vertex list no such matrix has been known: the bulk classification that produced them concerns closed chains, and most of the six are not PT symmetric, so even the ordering of the reflection equation is part of the problem. We classify, for each of the six and at every point of its full complex parameter variety, every single-valued meromorphic 2×2 reflection matrix that is holomorphic at the origin and proportional to the identity there. The classification starts from four unrestricted meromorphic entries: one differentiation of the reflection equation at the origin produces a homogeneous linear system of sixteen equations in those entries, and an exact stratification of each parameter variety by the rank of that system splits it into finitely many cells carrying either a unique kernel or a family of arbitrary functions fixed by a centralizer. The tables include every removable limit, singular stratum, arbitrary-function fiber and overlap, and every family continues to a single-valued meromorphic function on the plane. Four of the six classes admit no crossing datum whatsoever, because the first-leg partial transpose of their bulk matrices has a zero row and a zero column; the remaining two admit a unique one exactly where the rate parameter is nonzero, but the left boundary that datum prescribes yields commuting double-row transfer matrices for every independently chosen pair of boundaries only on the rational cells. Fixing the left boundary to the identity instead removes that restriction for Class 6 entirely: every classified right boundary, at every point of the Class-6 parameter plane and for every chain length, gives a commuting double-row family whose logarithmic derivative at the origin is twice the bulk Hamiltonian plus a one-site endpoint term set by the derivative of the boundary matrix.

1. INTRODUCTION

A quantum spin chain is integrable when it carries a tower of mutually commuting conserved charges, one of which is its Hamiltonian, and all of which are generated by a single object: an invertible matrix depending on a complex spectral parameter and solving the Yang–Baxter equation, the identity guaranteeing that multi-particle scattering factorizes into two-particle events [1–3]. Because everything else follows from that one matrix, the natural question at a fixed local dimension is not which models are integrable but how many there are, and the answer has been pursued by direct classification since the constant case was settled [4–7]. For a chain of spins one-half, with the bulk matrix depending on its two rapidities only through their difference and reducing at coincident rapidities to the exchange of the two sites, the classification is now complete: fourteen independent solutions, of which eight are the familiar vertex models and six lie outside that list, several of them with nilpotent or non-diagonalizable Hamiltonian densities [8]. Those six have since been absorbed into a completed classification of all regular 4×4 solutions, which drops the difference-form assumption and adds four further models [9, 10]. The fifth of them has been analysed as a closed chain whose transfer matrix and Hamiltonian are both non-diagonalizable, so that the algebraic Bethe ansatz does not reach the whole state space [11], and identified as a Jordanian Drinfeld twist of the Heisenberg chain whose continuum limit is a non-unitary deformation of the Landau–Lifshitz

Date: August 29, 2026.

2020 Mathematics Subject Classification. Primary 16T25; Secondary 81R12, 82B23, 13P10.

Key words and phrases. reflection equation, boundary Yang–Baxter equation, integrable spin chains, open chains, K -matrix, constructible classification.

model, with the sixth treated the same way [12]. The classification method itself has been carried to electron chains with a four-dimensional local space [13], and to the nearest-neighbour interactions underlying deformations of the $AdS_5 \times S^5$ worldsheet model [14]. The commuting charges these matrices generate are organized by a Yangian, whose representation theory is the structural background to the whole programme [15].

Everything in that list concerns a ring. Cutting the ring open at a wall requires one further piece of algebraic data: a reflection matrix, giving the amplitude with which an excitation returns from the wall with its rapidity reversed, and solving the reflection equation — the statement that two excitations may bounce off the wall in either order with the same total amplitude [16–18]. That equation is what preserves the commuting tower after the cut, and its solutions are what turn a list of bulk models into a catalogue of integrable boundary interactions; in algebraic language they select the part of the bulk symmetry that survives at the wall, a coideal subalgebra [19–21]. For the standard chains the reflection matrices are known completely [22], and for rational reflection matrices that stay invertible as the spectral parameter grows, the possible residual symmetry algebras have been derived from the boundary Yang–Baxter equation alone [23]. Boundaries have also been found for individual deformed models, the Jordanian open XXX chain among them [24], and integrable open ends are actively being constructed for discrete-time circuits built from the same algebraic data [25, 26]. For the six families outside the vertex list there is nothing: no reflection solution, no crossing relation, and no open-chain Hamiltonian. The paper that isolated them names precisely this as future work [8].

The nearest thing to a precedent is at a different local dimension. The same differential approach has been pushed to spin-one vertex models: for the fifteen-vertex R -matrices obeying the ice rule, which are 9×9 and not symmetric, four families of regular bulk matrices were found and their reflection matrices classified alongside, in the same ordering of the reflection equation used below [27]. That is the shape of result this paper supplies for the six 4×4 difference-form families, which are a different list of bulk matrices at a different local dimension.

Three features make the gap harder to close than it looks. First, most of the six bulk matrices are not PT symmetric, and for a bulk matrix that is not, the two orderings of the bulk factors in the reflection equation are genuinely different equations with genuinely different solution sets [28]; the literal equation is therefore part of the problem statement rather than a convention. Second, even the constant boundary — the identity matrix, which looks like no boundary at all — is a nontrivial condition, because substituting it leaves an identity between products of bulk matrices at shifted arguments that holds automatically only when the bulk matrix is symmetric. Third, the standard route from a right boundary to a left one runs through a crossing relation, the algebraic form of particle–antiparticle exchange, and nothing guarantees that these bulk matrices possess one. A list of checked examples would settle none of this: what is needed is a classification, over the whole of each complex parameter variety, including the singular strata where a parameter vanishes and a family degenerates.

This paper supplies that classification and its consequences. We fix the literal reflection equation, the class of admissible boundaries — single-valued meromorphic on the plane, holomorphic at the origin, and there a nonzero multiple of the identity — and the equivalence under which boundaries are counted, which is multiplication by a scalar function together with the constant similarities and spectral rescalings that preserve the same literal bulk matrix. Every boundary is then found, at every bulk point, of all six classes. The derivation uses no ansatz: it begins with four unrestricted meromorphic entries, and a single differentiation of the reflection equation at the origin converts the problem into a homogeneous linear system of sixteen equations in those four entries, whose coefficients are the bulk matrix, its derivative, and the first derivative of the boundary at the origin. What then decides the answer is the rank of that system over the field of meromorphic functions, and the rank takes only three values. Where it is three the kernel is one-dimensional, so a single normalized matrix is the whole solution set. Where it is two or zero the equation collapses,

after an entire invertible change of variable, to the statement that the boundary commutes with itself at different rapidities, and the centralizer of a nonscalar 2×2 matrix is two-dimensional, so the solutions are exactly one arbitrary meromorphic function times one fixed traceless direction. The partition of each parameter variety into cells of constant rank is computed exactly, from radicals and associated primes of the ideals generated by the maximal minors, and it is that computation which carries the exhaustiveness of the tables.

The consequences for open chains split the six classes sharply. For four of them no crossing datum exists at any bulk point, and the reason is a single rank count: the first-leg partial transpose of each of those bulk matrices has a zero row and a zero column, so it cannot be inverted, while a crossing relation would force it to be. For the remaining two, crossing exists exactly where the rate parameter is nonzero and is then unique up to scale; but the left boundary it prescribes produces commuting double-row transfer matrices for every independently chosen pair of right boundaries only on the rational cells, and we give the explicit nonzero commutator entries that rule out the rest.

Requiring every independently chosen pair to work is, however, more than an open chain needs. One compatible left boundary per right boundary already gives a commuting family, and for Class 6 there is one, the same for all of them: the identity. For every point of the Class-6 parameter plane, every classified right boundary, and every chain length, the double-row transfer matrices built with the identity on the left commute, and their logarithmic derivative at the origin is twice the sum of the bulk two-site densities plus a single one-site term at the end carrying the boundary, with no contribution from the auxiliary end because both partial traces of the Class-6 density vanish. This is proved on the generic stratum from the dual reflection identity that the boundary transfer-matrix theorem requires [29], extended to the singular stratum by polynomial continuation in the rate, and settled directly at the point where the bulk matrix degenerates to the exchange operator. No analogous construction is claimed for the other five classes.

Section 2 fixes the conventions, prints the six bulk matrices, and states the question. Section 3 derives the differentiated system and proves the rank trichotomy that makes it exhaustive. Section 4 gives the classification, class by class. Section 5 carries out the fixed-bulk quotient. Section 6 settles crossing, left boundaries, and the resulting open chains. Section 7 proves the Class-6 construction. Section 8 reports the independent checks and says what each does and does not establish, and Section 9 states the limitations.

2. SETTING

2.1. Bulk and boundary data. Let $V = \mathbb{C}^2$ be the state space of one site. In tensor-leg notation a subscript records which factors of a tensor power an operator acts on, so that R_{12} acts on the first two factors of $V^{\otimes 3}$ and as the identity on the third. A bulk matrix of difference form is an invertible $R(u) \in \text{End}(V \otimes V)$, depending on one complex spectral parameter, satisfying

$$R_{12}(u-v)R_{13}(u)R_{23}(v) = R_{23}(v)R_{13}(u)R_{12}(u-v). \quad (2.1)$$

Reading the two sides as amplitudes for three world lines, Eq. (2.1) says that a third line may be slid through the crossing of the other two without changing the amplitude, which is what makes multi-particle scattering factorize.

Write $P(x \otimes y) = y \otimes x$ for the exchange of the two factors and $R_{21} = PR_{12}P$ for the bulk matrix with its two legs interchanged. The bulk matrix is *regular* when it is holomorphic at the origin with

$$R(0) = P, \quad (2.2)$$

so that at coincident rapidities the two excitations simply swap places; every matrix considered below is regular. Regularity makes the logarithmic derivative of the closed-chain transfer matrix

local, and the two-site density it produces is

$$h = PR'(0). \quad (2.3)$$

The matrix h is the interaction of one nearest-neighbour bond: the closed-chain Hamiltonian is the sum of its copies over the bonds of the ring.

A bulk matrix is PT symmetric when swapping its two spaces has the same effect as transposing both, $R_{21}(u) = R_{12}^{t_1 t_2}(u)$, with t_1, t_2 the transposition on the first or second leg. The six- and eight-vertex matrices are PT symmetric; most of the six families below are not, and that is why the ordering of the bulk factors in the next equation has to be stated rather than assumed.

An integrable reflecting end is a matrix-valued function $K(u)$ on V : an excitation arriving with rapidity u leaves with rapidity $-u$ and its internal state multiplied by $K(u)$. Throughout this paper the right reflection equation is the literal identity

$$R_{12}(u-v)K_1(u)R_{21}(u+v)K_2(v) = K_2(v)R_{12}(u+v)K_1(u)R_{21}(u-v), \quad (2.4)$$

with $K_1 = K \otimes I_2$ and $K_2 = I_2 \otimes K$. Its content is that two excitations may reflect off the wall in either order, with the appropriate bulk exchanges in between, and give the same amplitude. Interchanging the roles of R_{12} and R_{21} in Eq. (2.4) gives a different equation whenever the bulk matrix is not PT symmetric, and in general a different solution set [28]; Eq. (2.4) is the equation solved here.

Definition 2.1 (Admissible boundary). An admissible boundary for a given bulk matrix is a single-valued meromorphic function $K : \mathbb{C} \rightarrow \text{Mat}_2(\mathbb{C})$, holomorphic at the origin, with $K(0) = \kappa I_2$ for some $\kappa \neq 0$, satisfying Eq. (2.4) as an identity of meromorphic functions on $\mathbb{C}^2$. No invertibility is required away from the origin.

The normalization at the origin is what makes a boundary an endpoint of the same chain the bulk matrix describes; the meromorphy is global, so a solution defined only near the origin, or only on a sheet of a branched surface, does not count until it is shown to continue.

Two admissible boundaries describe the same endpoint when they differ by the following equivalence. Multiplication by a nonzero scalar meromorphic function holomorphic and nonzero at the origin is a gauge; it can always be used to normalize $K(0) = I_2$, and it is used that way from here on. At a fixed bulk matrix the only further identification is

$$K(u) \sim f(u) S K(\lambda u) S^{-1}, \quad (2.5)$$

where f is such a scalar gauge, $\lambda \in \mathbb{C}^\times$, $S \in GL_2(\mathbb{C})$, and the pair (λ, S) preserves the same literal bulk matrix projectively:

$$(S \otimes S) R(\lambda u) (S^{-1} \otimes S^{-1}) = \varphi(u) R(u), \quad \varphi(0) = 1, \quad (2.6)$$

for some nonzero meromorphic φ . Written out, Eq. (2.6) says that rescaling the rapidity and conjugating both legs returns the same bulk matrix up to an overall function, so that the chain has not been changed. A similarity that moves the bulk parameters, that is singular on a degeneration, or that depends on the rapidity is therefore not an equivalence, however natural it may look.

2.2. The six bulk matrices. Fix the ordered basis $(e_1 \otimes e_1, e_1 \otimes e_2, e_2 \otimes e_1, e_2 \otimes e_2)$ of $V \otimes V$, in which

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (2.7)$$

This is the exchange operator written out: it moves the second basis vector of the middle block to the third and back, and fixes the two aligned states.

The six families are the ungraded Classes 1–6 of [8], quoted here from version 3 of that reference. All parameters are complex, and every apparent parameter singularity is resolved by analytic continuation. Class 1 is subject to $a_1 a_3 = a_2 a_4$ and reads

$$R_1(u) = \begin{pmatrix} 1 & \frac{a_1(e^{a_5 u} - 1)}{a_5} & \frac{a_2(1 - e^{-a_5 u})}{a_5} & \frac{(a_1 a_3 + a_2 a_4)(\cosh(a_5 u) - 1)}{a_5^2} \\ 0 & 0 & e^{-a_5 u} & \frac{a_4(1 - e^{-a_5 u})}{a_5} \\ 0 & e^{a_5 u} & 0 & \frac{a_3(e^{a_5 u} - 1)}{a_5} \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (2.8)$$

Each quotient in Eq. (2.8) is read at $a_5 = 0$ as its limit, so that the upper right entry becomes $(a_1 a_3 + a_2 a_4)u^2/2$ and the exponentials become 1; a_5 is the rate at which the two off-diagonal amplitudes grow.

For Class 2 put

$$\mathcal{N} = \begin{pmatrix} 0 & a_2 & a_3 - a_2 & a_5 \\ 0 & a_1 & 0 & a_4 \\ 0 & 0 & -a_1 & a_3 - a_4 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (2.9)$$

a nilpotent-on-the-diagonal generator whose square is strictly upper triangular, and set

$$R_2(u) = uP \left[\frac{a_1}{\sinh(a_1 u)} I_4 + \mathcal{N} + \frac{\tanh(a_1 u/2)}{a_1} \mathcal{N}^2 \right]. \quad (2.10)$$

The bulk matrix of Class 2 is thus the exchange operator dressed by a quadratic polynomial in one fixed matrix, with rapidity-dependent coefficients; at $a_1 = 0$ it is $P[I_4 + u\mathcal{N} + u^2\mathcal{N}^2/2]$. The source prints the leading scalar without an explicit I_4 ; multiplying by I_4 is the dimensionally consistent reading and the only one compatible with $R_2(0) = P$.

Class 3 is

$$R_3(u) = \begin{pmatrix} e^{-a_1 u} & a_3(e^{(a_1 - a_2)u} - e^{-a_1 u}) & a_3(e^{(a_1 + a_2)u} - e^{-a_1 u}) & 0 \\ 0 & 0 & e^{(a_1 + a_2)u} & 0 \\ 0 & e^{(a_1 - a_2)u} & 0 & 0 \\ 0 & 0 & 0 & e^{-a_1 u} \end{pmatrix}, \quad (2.11)$$

a six-vertex-like pattern in which the two middle amplitudes carry independent exponential rates and a_3 couples the aligned state to the two mixed ones. Class 4 is

$$R_4(u) = \begin{pmatrix} e^{a_1 u} & \frac{a_2 \sinh(a_1 u)}{a_1} & \frac{a_2 \sinh(a_1 u)}{a_1} & \frac{e^{a_1 u}(a_2 a_4 + a_1 a_3 \coth(a_1 u)) \sinh^2(a_1 u)}{a_1^2} \\ 0 & 0 & e^{-a_1 u} & \frac{a_4 \sinh(a_1 u)}{a_1} \\ 0 & e^{-a_1 u} & 0 & \frac{a_4 \sinh(a_1 u)}{a_1} \\ 0 & 0 & 0 & e^{a_1 u} \end{pmatrix}; \quad (2.12)$$

its two off-diagonal columns are equal, which is the structural feature that later forces a conic rather than a line of admissible first derivatives. At $a_1 = 0$ its rows are $(1, a_2 u, a_2 u, a_3 u + a_2 a_4 u^2)$, $(0, 0, 1, a_4 u)$, $(0, 1, 0, a_4 u)$ and $(0, 0, 0, 1)$.

Class 5 is

$$R_5(u) = (1 - a_1 u) \begin{pmatrix} 1 + 2a_1 u & a_2 u & -a_2 u & a_2 a_3 u^2 \\ 0 & 2a_1 u & 1 & -a_3 u \\ 0 & 1 & 2a_1 u & a_3 u \\ 0 & 0 & 0 & 1 + 2a_1 u \end{pmatrix}, \quad (2.13)$$

TABLE 1. The bulk parameter varieties, with the loci a complete boundary table must cover in addition to the generic point.

Class	Parameter variety	Loci that must be treated separately
1	$\mathcal{A}_1 = \{\mathbf{a} \in \mathbb{C}^5 : a_1 a_3 = a_2 a_4\}$	the analytic $a_5 = 0$ limit
2	$\mathcal{A}_2 = \mathbb{C}^5$	the analytic $a_1 = 0$ limit
3	$\mathcal{A}_3 = \mathbb{C}^3$	$a_3 = 0$, and every coincidence of exponential rates
4	$\mathcal{A}_4 = \mathbb{C}^4$	the analytic $a_1 = 0$ limit, and the degenerate conic
5	$\mathcal{A}_5 = \mathbb{C}^3$	$a_2 + a_3 = 0$, and the origin
6	$\mathcal{A}_6 = \mathbb{C}^2$	$a_2 = 0$, and the origin

and Class 6 is

$$R_6(u) = (1 - a_1 u)(1 + 2a_1 u) \begin{pmatrix} 1 & a_2 u & a_2 u & -a_2^2 u^2 (1 + 2a_1 u) \\ 0 & \frac{2a_1 u}{1+2a_1 u} & \frac{1}{1+2a_1 u} & -a_2 u \\ 0 & \frac{1}{1+2a_1 u} & \frac{2a_1 u}{1+2a_1 u} & -a_2 u \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (2.14)$$

These last two are rational rather than exponential: every entry is a polynomial in the rapidity after the prefactor of Eq. (2.14) cancels the apparent pole of its middle block. Each of Eqs. (2.8) and (2.10) to (2.14) satisfies $R_j(0) = P$, so all six are regular and each defines a closed chain with a two-site density Eq. (2.3).

The parameter varieties over which the classification runs are those of Table 1, each carrying the loci at which a family degenerates.

2.3. The question, and why a list of examples cannot answer it. For each $j \in \{1, \dots, 6\}$ and each $\mathbf{a} \in \mathcal{A}_j$, determine every admissible boundary in the sense of Definition 2.1, together with all overlaps and degenerations between the resulting families and their orbits under Eq. (2.5). The quantifier includes every singular and removable bulk locus.

That the trivial-looking boundary is already a constraint is seen by substituting it. With $K(u) = I_2$, Eq. (2.4) becomes

$$R_{12}(u - v)R_{21}(u + v) = R_{12}(u + v)R_{21}(u - v), \quad (2.15)$$

an identity between products of bulk matrices at shifted arguments which is automatic for a PT -symmetric bulk matrix and not otherwise. So even the constant boundary has to be checked class by class; and a list of boundaries that pass the check, however long, still says nothing about the ones not on it.

At the opposite extreme sits a bulk point at which the equation says almost nothing, and it is instructive because it is the source of every arbitrary-function family below.

Example 2.2 (The constant fiber). If $R(u) = P$ identically, then $R_{21} = P$ as well and Eq. (2.4) collapses to

$$I_2 \otimes (K(u)K(v) - K(v)K(u)) = 0. \quad (2.16)$$

The reflection equation has become the statement that the boundary matrix commutes with itself at any two rapidities, and nothing more.

Lemma 2.3 (Centralizer). *Suppose $K : \mathbb{C} \rightarrow \text{Mat}_2(\mathbb{C})$ is meromorphic, that $K(0) = I_2$, and that $[K(u), K(v)] = 0$ holds for all u and v . Then either K is scalar, or there is a fixed nonzero traceless*

TABLE 2. Symbols in force throughout. Class-specific abbreviations are declared where the class is treated.

Symbol	Meaning
$V, P, R_j(u)$	site space $\mathbb{C}^2$; exchange operator Eq. (2.7) ; the bulk matrix of class j
$\mathcal{A}_j, \mathbf{a}$	the bulk parameter variety of class j , and a point of it
h	the two-site density $PR'(0)$
$K(u), \kappa$	right boundary, and its value $K(0) = \kappa I_2$ at the origin
k_1, k_2, k_3, k_4	the entries of $K(u)$, read row by row
B, b, c, d	the traceless first derivative of K at the origin, and its entries Eq. (3.1)
$f(u), \lambda, S, \varphi(u)$	the scalar gauge, spectral scale, similarity and bulk scalar of Eqs. (2.5) and (2.6)
$\psi(u), B_0$	an arbitrary meromorphic function vanishing at 0, and a fixed traceless direction
$E_{ij}, \sigma_x, \sigma_y, \sigma_z$	the matrix units and the Pauli matrices
$\mathcal{C}_{R,B}(u)$	the 16×4 coefficient matrix Eq. (3.3)
$M, \rho, \chi(u)$	the crossing matrix, shift and scalar Eq. (6.1)
$L, T_a, \widehat{T}_a, t_L(u)$	chain length, the two monodromies, and the double-row transfer matrix
$K_L, \mathcal{H}_L$	the left boundary and the open-chain Hamiltonian
$\Lambda(u), \Gamma(u)$	a self-commuting family and the entire twist conjugating it

$B_0 \in \text{Mat}_2(\mathbb{C})$ and a meromorphic ψ with $\psi(0) = 0$ such that, modulo the scalar gauge,

$$K(u) = I_2 + \psi(u)B_0. \quad (2.17)$$

Conversely every such K commutes with itself at all rapidities.

Proof. If $K(u_0)$ is nonscalar for some u_0 , then every $K(v)$ lies in the centralizer of $K(u_0)$, which for a 2×2 matrix is the two-dimensional span of I_2 and $K(u_0)$. Hence $K(v) = \alpha(v)I_2 + \beta(v)K(u_0)$ with meromorphic coefficients, and subtracting the trace and dividing by the scalar gauge gives [Eq. \(2.17\)](#) with B_0 the traceless part of $K(u_0)$. Two matrices of that form commute because B_0 commutes with itself. $\square$

[Equation \(2.17\)](#) is load-bearing rather than decorative: the free function ψ is an arbitrary meromorphic function vanishing at the origin, so a solution may have its first nonzero Taylor coefficient at any order whatever. No polynomial ansatz, and no truncation of the power series at a fixed order, can see all of these.

[Table 2](#) collects the symbols that recur throughout the paper; those belonging to a single class are declared where that class is treated.

3. THE DIFFERENTIATED SYSTEM AND WHY IT IS EXHAUSTIVE

3.1. From the reflection equation to a linear system. Take a representative normalized by $K(0) = I_2$ and write its entries and its traceless first derivative as

$$K(u) = \begin{pmatrix} k_1(u) & k_2(u) \\ k_3(u) & k_4(u) \end{pmatrix}, \quad B := K'(0) - \frac{1}{2} \text{tr}(K'(0))I_2 = \begin{pmatrix} b & c \\ d & -b \end{pmatrix}. \quad (3.1)$$

The four functions $k_1, \dots, k_4$ are unrestricted meromorphic functions: no entry is set to zero and no diagonal, triangular, polynomial, rational or exponential form is assumed. The matrix B is the only datum of the boundary that survives the scalar gauge at first order, and (b, c, d) is called the *slope* of the boundary. For the derivation the remaining scalar freedom is used to impose $\text{tr} K(u) = 2$

near the origin, which is possible because $\text{tr } K(0) = 2$; then $K'(0) = B$ literally. The displayed representatives in [Section 4](#) sometimes use the more convenient gauge $k_1 = 1$ instead, and their traceless first derivative is still [Eq. \(3.1\)](#).

Differentiate [Eq. \(2.4\)](#) with respect to v at $v = 0$, using $K(0) = I_2$ and $K'(0) = B$. Both sides become linear in the single remaining unknown function $K(u)$, and collecting the coefficient of each entry of $K(u)$ gives a linear map. Explicitly, for a matrix unit E_{pq} ,

$$\begin{aligned} \Phi_{R,B}(E_{pq}) = & -2R'(u)(E_{pq} \otimes I_2)R_{21}(u) + 2R(u)(E_{pq} \otimes I_2)R'_{21}(u) \\ & + R(u)(E_{pq} \otimes I_2)R_{21}(u)B_2 - B_2R(u)(E_{pq} \otimes I_2)R_{21}(u), \end{aligned} \quad (3.2)$$

where $B_2 = I_2 \otimes B$. Each of the four terms of [Eq. \(3.2\)](#) is the contribution of one differentiated factor of [Eq. \(2.4\)](#): the two bulk matrices at shifted arguments contribute the derivative terms with opposite signs, and the boundary at the second leg contributes the commutator with B . Vectorizing $\Phi_{R,B}(E_{pq}) \in \text{Mat}_4(\mathbb{C})$ row by row gives a column of length sixteen; assembling the four columns in the order $(E_{11}, E_{12}, E_{21}, E_{22})$ gives a 16×4 matrix $\mathcal{C}_{R,B}(u)$, and every admissible boundary obeys

$$\mathcal{C}_{R,B}(u) (k_1(u), k_2(u), k_3(u), k_4(u))^t = 0. \quad (3.3)$$

In words: at each rapidity the four entries of an admissible boundary form a vector in the kernel of an explicitly computable 16×4 matrix built from the bulk data and the boundary slope. [Equation \(3.3\)](#) is necessary, not sufficient; sufficiency is restored below by substituting each candidate back into [Eq. \(2.4\)](#). A scalar factor in R multiplies [Eq. \(3.2\)](#) by its square and the two mixed terms cancel, so the kernel is unchanged by the removal of an overall scalar from the bulk matrix; the scalars are restored in every direct verification and in every open-chain computation.

3.2. The rank trichotomy. Over the field of meromorphic functions, $\mathcal{C}_{R,B}(u)$ has a rank, and that rank is what decides the answer.

Lemma 3.1 (Rank trichotomy). *Fix j and a bulk point $\mathbf{a} \in \mathcal{A}_j$, and a slope (b, c, d) . Then $\text{rank } \mathcal{C}_{R,B}$ over the meromorphic function field is 3, 2, or 0.*

- (1) *If the rank is three, the kernel is one-dimensional. The unique normalized generator, if it satisfies [Eq. \(2.4\)](#), is the whole set of admissible boundaries with that slope, modulo the scalar gauge; if it does not, there is none.*
- (2) *If the rank is two or zero, the equations obtained by specializing [Eq. \(2.4\)](#) to that cell — computed there directly, never by dividing through by a quantity that vanishes on it — reduce, after conjugation by an entire and everywhere invertible twist $\Gamma(u)$, to $[\Lambda(u), \Lambda(v)] = 0$ for $K(u) = \Gamma(u)\Lambda(u)\Gamma(u)$. By [Lemma 2.3](#) the admissible boundaries on that cell are exactly $\Gamma(I_2 + \psi B_0)\Gamma$ for a fixed traceless direction B_0 and an arbitrary meromorphic ψ with $\psi(0) = 0$, together with the scalar branch.*

The lemma is decided cell by cell, and the cells are the content of the next statement. It is the one step of this paper carried by exhaustive symbolic computation, so we say exactly what was computed and why it settles the question.

Lemma 3.2 (Constructible stratification). *For each j , the product of the bulk parameter variety $\mathcal{A}_j$ with the slope space $\mathbb{C}^3$ is partitioned into finitely many disjoint constructible cells, listed in [Section 4](#), on each of which $\text{rank } \mathcal{C}_{R,B}$ is constant. The cells cover the product exactly; no residual component remains.*

Proof, by exhaustive symbolic computation. The object attached to a point of $\mathcal{A}_j \times \mathbb{C}^3$ is the explicit 16×4 matrix $\mathcal{C}_{R,B}(u)$ of [Eq. \(3.3\)](#), whose entries are polynomials in the bulk parameters, the slope, and — depending on the class — either the rapidity or the exponentials $e^{\lambda u}$ appearing in the bulk matrix. The test applied to a point is the rank of that matrix, and the rank is decided by which of its minors vanish identically in u . This is decisive because [Lemma 3.1](#) turns each of the three

possible ranks into a complete description of the solution set at that point: at rank three the kernel has dimension one and there is at most one boundary, at rank two or zero the equation is a commutator identity whose solutions [Lemma 2.3](#) exhausts. Nothing about the boundary beyond its slope enters the test.

The cells arise by construction, which is why they cover. Let I_4 be the ideal generated by the sixteen maximal minors of $\mathcal{C}_{R,B}$, regarded as polynomials in the bulk parameters and the slope after clearing the rapidity dependence. Its radical was computed exactly, together with the associated primes of that radical; the complement of $V(I_4)$ is the rank-three cell, and each associated prime gives a component on which the rank drops. On each such component the ideal of 3×3 minors was formed anew, from the matrix specialized to that component rather than from a generic basis, and its radical computed in turn; this is what distinguishes a genuine lower-rank stratum from an artefact of dividing by a quantity that vanishes there. The recursion terminates because each step strictly decreases dimension, and the resulting finitely many locally closed sets are disjoint after the precedence stated with each table. Two classes need one further refinement. For Class 3 the minors are exponential rather than polynomial, so a minor can vanish identically in u through a coincidence of exponential rates and not through the vanishing of a coefficient; every such coincidence was solved for, giving the finite list of rate collisions recorded in [Section 4.3](#), and each collision curve was treated as its own component with the matrix recomputed on it. For Class 4 the rank-drop locus is a conic in the slope, so it was saturated and covered by both projective charts, and the two points where the chart used for the generic formula degenerates were computed separately.

What follows is the classification of [Theorem 4.1](#): on each cell of the resulting partition, one of the two branches of [Lemma 3.1](#) applies, and its output is the family printed for that cell. $\square$

Some of the certificates are short enough to print, and they are the ones a reader is most likely to want to redo. In Class 2 with $a_1 \neq 0$, writing $\varepsilon = a_2 + a_4 - a_3$ and $\omega = a_3\varepsilon - a_1a_5$ and using the dimensionless slope, the radical of the maximal-minor ideal is

$$\sqrt{I_4} = \langle b(b+1)\varepsilon\omega, d\varepsilon, da_5 \rangle, \quad (3.4)$$

whose associated primes are $\langle b, d \rangle$, $\langle b+1, d \rangle$, $\langle \varepsilon, d \rangle$, $\langle \omega, d \rangle$ and $\langle \varepsilon, a_5 \rangle$. Read as a statement about boundaries, [Eq. \(3.4\)](#) says that at $a_1 \neq 0$ the rank can drop only when the lower left slope d vanishes together with one of two bulk discriminants or a distinguished value of b , or when the two bulk parameters ε and a_5 vanish together — and those five possibilities are exactly the five cells listed for this class in [Section 4.2](#). At $a_1 = 0$ the same ideal becomes $\langle ba_3\varepsilon, da_3\varepsilon \rangle = \langle b, d \rangle \cap \langle a_3 \rangle \cap \langle \varepsilon \rangle$. For Class 5 with $a \neq 0$ one maximal minor is

$$-2a[-u^2pd + u^2rd - 2ub - 2][-2u^2apd + 2u^2ard - upd + urd - 4a]^2, \quad (3.5)$$

whose constant term in u is $64a^3$; since that is nonzero whenever $a \neq 0$, the minor cannot vanish identically, so the rank is three for every slope and the Class-5 boundary set off $a = 0$ is a single family. At $a = 0$ with $p + r \neq 0$ the constant term of the corresponding row minor is $-8(p+r)^3$, giving the same conclusion there. For Class 6 with $a \neq 0$ the corresponding minor is

$$-64a^3(2u^3aQd - ub - 1), \quad (3.6)$$

again with nonzero constant term, so the rank is uniformly three. These four displays are the reason Classes 5 and 6 have single polynomial families off their degenerate loci while Classes 1–4 fragment.

3.3. Global continuation. The classification is a statement about meromorphic functions on the whole plane, not about germs, and the step from one to the other is the following.

Each isolated family below is a rational expression in u and in finitely many exponentials $e^{\lambda u}$, whose denominator is nonzero at $u = 0$ on the cell where it is stated. Such an expression is single-valued and meromorphic on all of $\mathbb{C}$; the later zeros of its denominator are poles, which [Definition 2.1](#) permits. On a rank-three cell, an arbitrary admissible boundary and the displayed

kernel generator agree on a nonempty open set of rapidities, hence everywhere, by the identity theorem for meromorphic functions. On a rank-two or rank-zero cell, the specialized equations determine the fixed traceless direction B_0 without dividing by any coordinate, so the resulting identity persists through the zeros and poles of the arbitrary function; and the twists Γ used there are entire and everywhere invertible, so conjugating by them neither creates nor destroys singularities. No branch choice, monodromy, or purely local continuation is hidden anywhere in the argument.

4. THE CLASSIFICATION

Throughout this section boundaries are normalized by $K(0) = I_2$ and listed modulo the scalar gauge, but before the fixed-bulk quotient of [Section 5](#). In every isolated family, (b, c, d) is the slope [Eq. \(3.1\)](#). Every function called ψ , ψ_1 or ψ_2 is an arbitrary single-valued meromorphic function on $\mathbb{C}$, holomorphic at the origin, and Λ denotes a family commuting with itself at all rapidities as in [Lemma 2.3](#).

Theorem 4.1 (Classification). *For every $j \in \{1, \dots, 6\}$ and every $\mathbf{a} \in \mathcal{A}_j$, the matrices listed in [Sections 4.1 to 4.6](#), and no others, are the admissible boundaries of [Definition 2.1](#) for R_j at $\mathbf{a}$. The cells are finite in number, constructible, disjoint after the stated precedence, and cover the full parameter variety. Each family of arbitrary functions contains its own finite specializations, and the overlap statements say where two families meet.*

The proof is given after the six subsections, once the objects it refers to have been displayed.

4.1. **Class 1.** Assume $a_1 a_3 = a_2 a_4$, write $\theta = e^{a_5 u}$ for the bulk exponential, and set

$$\mathcal{E}_1 = \{a_1 = a_4, a_2 = a_3\}. \quad (4.1)$$

The locus $\mathcal{E}_1$ is where the two off-diagonal columns of R_1 acquire matching rates, and it is the only Class-1 locus carrying arbitrary functions at $a_5 \neq 0$. For $a_5 \neq 0$ define

$$K_+(u) = \begin{pmatrix} 1 + \frac{b}{a_5}(1 - \theta^{-1}) & \frac{c}{2a_5}(\theta - \theta^{-1}) \\ 0 & 1 - \frac{b}{a_5}(\theta - 1) \end{pmatrix}, \quad (4.2)$$

$$K_-(u) = \begin{pmatrix} \theta^{-2} \frac{(b+2a_5)\theta - b - a_5}{a_5} & 0 \\ \frac{d}{2a_5}(\theta - \theta^{-1}) & \theta \frac{-(b+a_5)\theta + b + 2a_5}{a_5} \end{pmatrix}. \quad (4.3)$$

These are the upper- and lower-triangular boundaries whose off-diagonal entry grows like the bulk amplitude it must follow; each is determined by its slope alone. On $\mathcal{E}_1$ let

$$\Gamma_1(u) = \theta^{-1/2} \begin{pmatrix} 1 & -\frac{a_2}{a_5}(\theta - 1) \\ 0 & \theta \end{pmatrix}, \quad (4.4)$$

an entire and everywhere invertible matrix function, since $\theta^{-1/2} = e^{-a_5 u/2}$ and the determinant is 1. The complete family there is

$$K(u) = \Gamma_1(u) \Lambda(u) \Gamma_1(u), \quad [\Lambda(u), \Lambda(v)] = 0, \quad (4.5)$$

so on $\mathcal{E}_1$ the reflection equation says only that a twisted version of the boundary commutes with itself, and every self-commuting family is allowed. The traceless slope of Λ has coordinates $(b + a_5, c + 2a_2, d)$.

Off $\mathcal{E}_1$ the remaining cells are those of [Table 3](#).

For $a_5 = 0$ the exponentials degenerate and the boundary becomes polynomial:

$$K_0(u) = \begin{pmatrix} 1 + bu - \frac{a_2 + a_3}{2} du^2 & cu + a_2 a_3 du^3 \\ du & 1 - bu - \frac{a_2 + a_3}{2} du^2 \end{pmatrix}, \quad (4.6)$$

TABLE 3. Class 1 at $a_5 \neq 0$, off the locus $\mathcal{E}_1$.

Bulk and slope cell	Complete family
$(a_2, a_3) \neq (0, 0)$ and $(a_1, a_4) \neq (0, 0)$	$d = 0$ forced; uniquely K_+
$(a_2, a_3) \neq (0, 0)$, $(a_1, a_4) = (0, 0)$, $b \neq -a_5$	$d = 0$ forced; uniquely K_+
the same bulk cell with $b = -a_5$	$\psi_1(u) \text{diag}(1, e^{2a_5 u}) + \psi_2(u)E_{12}$, with $\psi_1(0) = 1$, $\psi_2(0) = 0$, $\psi'_1(0) = -a_5$, $\psi'_2(0) = c$
$(a_2, a_3) = (0, 0)$, $d = 0$, $c \neq 0$	uniquely K_+
$(a_2, a_3) = (0, 0)$, $c = 0$, $d \neq 0$	uniquely K_-
$(a_2, a_3) = (0, 0)$, $c = d = 0$	every normalized diagonal family, with traceless slope $\text{diag}(b, -b)$

TABLE 4. Class 2 at $a_1 \neq 0$.

Bulk and slope cell	Complete family
$\varepsilon = 0$, $a_5 = 0$	the twisted commuting family Eq. (4.12); every slope
$\varepsilon = 0$, $a_5 \neq 0$, $b \neq -a_1$	$d = 0$ forced; uniquely K_ε
$\varepsilon = 0$, $a_5 \neq 0$, $b = -a_1$	$\begin{pmatrix} e^{-a_1 u} & \psi(u) \\ 0 & e^{a_1 u} \end{pmatrix}$, $\psi(0) = 0$, $\psi'(0) = c$
$\varepsilon \neq 0$, $\omega = 0$	$d = 0$ forced; uniquely K_ω , for arbitrary b, c
$\varepsilon \omega \neq 0$	$d = 0$ and $b \in \{0, -a_1\}$ forced; uniquely K_ω

a cubic in the rapidity whose higher coefficients are fixed by the slope and the bulk parameters. Every slope occurs. Off $\mathcal{E}_1$, Eq. (4.6) is the unique family except in two cases: at $(a_1, a_4) = (0, 0)$ with $(a_2, a_3) \neq (0, 0)$ and $b = d = 0$, every $\psi_1 I_2 + \psi_2 E_{12}$ with $\psi_1(0) = 1$, $\psi_2(0) = 0$, $\psi'_1(0) = 0$, $\psi'_2(0) = c$ occurs; and at $(a_2, a_3) = (0, 0)$ with $(a_1, a_4) \neq (0, 0)$ and $c = d = 0$, every normalized diagonal family occurs. On $\mathcal{E}_1$, with $\Gamma_1(u) = I_2 - a_2 u E_{12}$, the complete family is again

$$K(u) = \Gamma_1(u) \Lambda(u) \Gamma_1(u), \quad [\Lambda(u), \Lambda(v)] = 0, \quad (4.7)$$

the $a_5 \rightarrow 0$ limit of Eq. (4.5), with auxiliary slope $(b, c + 2a_2, d)$. The finite formulas above are contained in these arbitrary-function families wherever the two overlap, so no boundary is counted twice.

4.2. Class 2. Set

$$\varepsilon = a_2 + a_4 - a_3, \quad \omega = a_3 \varepsilon - a_1 a_5, \quad (4.8)$$

the two discriminants that Eq. (3.4) showed to control the rank drops: ε measures the failure of the three linear parameters to balance, and ω couples that failure to the quadratic parameter a_5 . For $a_1 \neq 0$ define

$$K_\omega(u) = \begin{pmatrix} 1 + \frac{b}{a_1}(1 - e^{-a_1 u}) & \frac{c}{a_1} \sinh(a_1 u) \\ 0 & 1 - \frac{b}{a_1}(e^{a_1 u} - 1) \end{pmatrix}, \quad (4.9)$$

$$K_\varepsilon(u) = \begin{pmatrix} \cosh(a_1 u) + \frac{b}{a_1} \sinh(a_1 u) & \frac{\sinh(a_1 u)}{a_1^2} \{ (a_1 c - 2a_4 b) \cosh(a_1 u) + 2a_4 b \} \\ 0 & \cosh(a_1 u) - \frac{b}{a_1} \sinh(a_1 u) \end{pmatrix}. \quad (4.10)$$

Both are upper triangular with unit determinant to leading order, and they differ in which discriminant vanishes: K_ω is the boundary on $\omega = 0$ and K_ε the boundary on $\varepsilon = 0$. The nonzero-rate partition is Table 4.

TABLE 5. Class 2 at $a_1 = 0$.

Cell	Complete family
$\varepsilon = 0, a_5 = 0$	$(I_2 - a_4 u E_{12}) \Lambda(u) (I_2 - a_4 u E_{12})$ with $[\Lambda(u), \Lambda(v)] = 0$; auxiliary slope $(b, c + 2a_4, d)$
$\varepsilon = 0, a_5 \neq 0, (b, d) \neq (0, 0)$	uniquely $K_{\varepsilon, 0}$; c arbitrary
$\varepsilon = 0, a_5 \neq 0, b = d = 0$	$I_2 + \psi(u) E_{12}$, $\psi(0) = 0$, $\psi'(0) = c$
$a_3 = 0, \varepsilon \neq 0$	uniquely $K_{a_3, 0}$; every slope
$a_3 \varepsilon \neq 0$	uniquely $I_2 + cu E_{12}$

On the first row of Table 4 put

$$\Xi = \begin{pmatrix} 0 & a_4 \\ 0 & -a_1 \end{pmatrix}, \quad \Gamma_2(u) = e^{-a_1 u/2} e^{-u\Xi}, \quad (4.11)$$

again entire and everywhere invertible, and then

$$K(u) = \Gamma_2(u) \Lambda(u) \Gamma_2(u), \quad [\Lambda(u), \Lambda(v)] = 0, \quad (4.12)$$

with auxiliary slope $(b + a_1, c + 2a_4, d)$: on that cell the bulk matrix factorizes through the exchange operator and an entire twist, so the reflection equation again reduces to self-commutation.

For $a_1 = 0$ define the two polynomial boundaries

$$K_{\varepsilon, 0}(u) = \begin{pmatrix} 1 + bu - a_4 du^2 & cu + a_4^2 du^3 \\ du & 1 - bu - a_4 du^2 \end{pmatrix}, \quad (4.13)$$

$$K_{a_3, 0}(u) = \begin{pmatrix} 1 + bu + \frac{a_2 - a_4}{2} du^2 & cu - a_2 a_4 du^3 \\ du & 1 - bu + \frac{a_2 - a_4}{2} du^2 \end{pmatrix}, \quad (4.14)$$

whose quadratic and cubic coefficients differ exactly by which bulk combination survives the limit. The zero-rate partition is Table 5.

4.3. Class 3. Class 3 is the class in which exponential rates rather than polynomial coefficients control the answer. Put

$$m = 2a_1 - a_2, \quad n = 2a_1 + a_2, \quad g = a_3, \quad X = e^{mu}, \quad Y = e^{nu}, \quad (4.15)$$

so that m and n are the two independent rates carried by the middle block and g is the coupling of the aligned state to the mixed ones. Removing the harmless overall scalar $e^{-a_1 u}$ leaves

$$\hat{R}_3(u) = \begin{pmatrix} 1 & g(X - 1) & g(Y - 1) & 0 \\ 0 & 0 & Y & 0 \\ 0 & X & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (4.16)$$

which shows that the whole class depends only on (m, n, g) , and that the bulk matrix is the exchange operator exactly when $m = n = 0$. The complete partition is Table 6, the $g \neq 0$ cells taking precedence over the $g = 0$ ones.

On the generic cell the two families are

$$K_D(u) = \text{diag} \left(1, \frac{(n - b)Y + b}{Y\{b(Y - 1) + n\}} \right) \quad (4.17)$$

TABLE 6. The complete constructible partition for Class 3.

Condition	Types of family that occur
$g \neq 0, mn(m-n)(m+n)(n+2m)(m+2n) \neq 0$	diagonal, and mixed upper
$g \neq 0, m = n \neq 0$	diagonal, lower, pure upper, mixed upper
$g \neq 0, n = -m \neq 0$	one full upper-triangular component
$g \neq 0, n = -2m \neq 0$	the inherited diagonal and mixed branches, plus one extra component
$g \neq 0, m = -2n \neq 0$	the inherited diagonal and mixed branches, plus one extra component
$g \neq 0, m = 0, n \neq 0$	unique, off one arbitrary-function stabilizer line
$g \neq 0, n = 0, m \neq 0$	unique, off an arbitrary diagonal stabilizer
$m = n = 0, \text{ any } g$	every self-commuting family
$g = 0, m + n \neq 0$	unique, off an arbitrary diagonal stabilizer
$g = 0, m = -n \neq 0$	arbitrary diagonal, plus a homogeneous projective family

and, for $d = 0$ and $b = -c/(2g)$,

$$K_U(u) = \begin{pmatrix} 1 & \frac{cg(X^2 - 1)}{2mgX - c(X - 1)} \\ 0 & \frac{X\{2mg + c(X - 1)\}}{2mgX - c(X - 1)} \end{pmatrix}, \quad (4.18)$$

with b free in Eq. (4.17) and c free in Eq. (4.18). The diagonal branch uses only the rate n and is available for every b ; the mixed branch exists only when the slope satisfies the one linear relation $2gb + c = 0$, which ties the upper entry of the boundary to the bulk coupling.

On $m = n = \mu \neq 0$ with $\theta = e^{\mu u}$, the diagonal and mixed branches are the specializations of Eqs. (4.17) and (4.18), and two further branches open:

$$K_{\text{lo}}(u) = \begin{pmatrix} 1 & 0 \\ \frac{d(\theta^2 - 1)}{gd(\theta - 1) + 2\mu\theta} & \frac{\theta\{2\mu - gd(\theta - 1)\}}{gd(\theta - 1) + 2\mu\theta} \end{pmatrix}, \quad c = 0, \quad b = \frac{gd}{2}, \quad (4.19)$$

$$K_{PU}(u) = I_2 + \frac{c(\theta^2 - 1)}{2\mu\theta} E_{12}, \quad b = d = 0. \quad (4.20)$$

When the two rates coincide the class regains the symmetry between the upper and the lower triangle, which is why a lower-triangular boundary exists there and nowhere else at $g \neq 0$.

The three cells on which the rates are in negative ratio are not optional, and they are the reason the exponential grouping of Lemma 3.2 is needed. With $\theta = e^{\mu u}$ and $n = -m \neq 0$, every slope with $d = 0$ and arbitrary b, c has the unique solution

$$K_{-1}(u) = \begin{pmatrix} 1 & \frac{c(\theta^2 - 1)}{2\{(m+b)\theta - b\}} \\ 0 & \frac{\theta\{m + b - b\theta\}}{(m+b)\theta - b} \end{pmatrix}. \quad (4.21)$$

If $n = -2m \neq 0$ the extra component is

$$d = 0, \quad b = -m, \quad K_{-2}(u) = \begin{pmatrix} 1 & c(\theta^2 - 1)/(2m) \\ 0 & \theta^2 \end{pmatrix}, \quad \theta = e^{\mu u}, \quad (4.22)$$

and if $m = -2n \neq 0$ it is

$$d = 0, \quad b = n, \quad K_{-1/2}(u) = \begin{pmatrix} 1 & c(\theta^2 - 1)/(2n\theta^2) \\ 0 & \theta^{-2} \end{pmatrix}, \quad \theta = e^{nu}. \quad (4.23)$$

Each of Eqs. (4.21) to (4.23) is a boundary that exists only on its own curve in the rate plane and is invisible from the generic formulas.

Remark 4.2. The necessity of these cells is not a formality. At $g = m = 1$ and $n = -1$ the matrix $I_2 + \sinh(u)E_{12}$ is an admissible boundary, and it is a specialization of Eq. (4.21) and of no generic-rate formula. A table built by solving the system once at generic rates and then substituting would omit it. Solving for the identical vanishing of the exponential minors shows that this happens for $n/m \in \{-2, -1, -\frac{1}{2}\}$ and for no other ratio, which is what makes the list of three complete.

On $g \neq 0, m = 0, n = \mu \neq 0$, put $\theta = e^{\mu u}$ and

$$\mathcal{D}_3 = 2\mu\theta + 2b\theta(\theta - 1) - gd(\theta - 1)^2. \quad (4.24)$$

Off the line $d = 0, b = -c/(2g)$ the unique boundary is

$$K_{m=0}(u) = \begin{pmatrix} 1 & \frac{c(\theta^2 - 1)}{\mathcal{D}_3} \\ \frac{d(\theta^2 - 1)}{\mathcal{D}_3} & \frac{2\mu\theta - 2b(\theta - 1) - gd(\theta - 1)^2}{\mathcal{D}_3} \end{pmatrix}, \quad (4.25)$$

the first boundary in this class with both off-diagonal entries nonzero, and on the excluded line the family is

$$K(u) \sim I_2 + \psi(u) \begin{pmatrix} 0 & 1 \\ 0 & 1/g \end{pmatrix}, \quad \psi(0) = 0, \quad \psi'(0) = c. \quad (4.26)$$

On $g \neq 0, n = 0, m = \mu \neq 0$, put $\theta = e^{\mu u}$ and $\mathcal{D}_4 = b(\theta - 1) + \mu\theta$; off $c = d = 0$,

$$K_{n=0}(u) = \begin{pmatrix} 1 & \frac{c(\theta^2 - 1)}{2\mathcal{D}_4} \\ \frac{d(\theta^2 - 1)}{2\mathcal{D}_4} & \frac{\theta\{\mu - b(\theta - 1)\}}{\mathcal{D}_4} \end{pmatrix}, \quad (4.27)$$

while on $c = d = 0$ every $\text{diag}(1, \psi(u))$ with $\psi(0) = 1$ and $\psi'(0) = -2b$ occurs. When $m = n = 0$, Eq. (4.16) is the exchange operator for every g , and Example 2.2 and Lemma 2.3 give the answer.

Two cells remain, both at $g = 0$. If $m + n \neq 0$, set $Z = XY = e^{(m+n)u}$ and $\mathcal{L} = (b + m)Z + n - b$. Off $c = d = 0$ the unique boundary is

$$K_{g=0}(u) = \begin{pmatrix} 1 & \frac{c(Z^2 - 1)}{2Y\mathcal{L}} \\ \frac{d(Z^2 - 1)}{2Y\mathcal{L}} & \frac{X\{(n - b)Z + b + m\}}{Y\mathcal{L}} \end{pmatrix}, \quad (4.28)$$

which depends on the two rates only through their sum in the numerators and through X/Y in the corner; on $c = d = 0$ every $\text{diag}(1, \psi(u))$ with $\psi(0) = 1$ and $\psi'(0) = -2b$ occurs. If instead $m = -n = \mu \neq 0$, put $\theta = e^{\mu u}$; besides the diagonal families the complete nondiagonal family is homogeneous,

$$K(u) \sim \begin{pmatrix} 1 & \gamma_1\psi(u) \\ \gamma_2\psi(u) & \theta^2 + \nu\theta\psi(u) \end{pmatrix}, \quad [\gamma_1 : \gamma_2] \in \mathbb{P}^1, \quad \psi(0) = 0, \quad (4.29)$$

a one-function family whose off-diagonal direction is a free point of the projective line rather than a fixed matrix. Its slope satisfies $c = \gamma_1\psi'(0)$, $d = \gamma_2\psi'(0)$ and $b = -\mu - \nu\psi'(0)/2$, and the presentation carries the redundancy $(\gamma_1, \gamma_2, \nu, \psi) \sim (\varsigma\gamma_1, \varsigma\gamma_2, \varsigma\nu, \psi/\varsigma)$ for $\varsigma \neq 0$. The chart $\gamma_1 = 0$,

TABLE 7. The complete constructible partition for Class 4.

Bulk condition	Types of family that occur
$a_1 \neq 0, \Sigma \neq 0, J \neq 0$	upper, mixed, the generic conic, and two vertical charts
$a_1 \neq 0, \Sigma \neq 0, J = 0$	upper, mixed, and one genuine degenerate line
$a_1 \neq 0, H = -A, G \neq A^2$	a single formula valid for every slope
$a_1 \neq 0, H = -A, G = A^2, A \neq 0$	that formula, plus an arbitrary-function stabilizer
$a_1 \neq 0, A = G = H = 0$	that formula, plus an arbitrary diagonal family
$a_1 = 0, a_2 \neq a_4$	a single formula valid for every slope
$a_1 = 0, a_2 = a_4, a_3 \neq 0$	that formula, plus an arbitrary upper family
$a_1 = 0, a_2 = a_4 \neq 0, a_3 = 0$	that formula, and twisted self-commuting families
$a_1 = a_2 = a_3 = a_4 = 0$	every self-commuting family

$\gamma_2 \neq 0$ is included explicitly, and no division by either off-diagonal coordinate is used, so the purely lower-triangular members are not lost.

4.4. **Class 4.** For $a_1 \neq 0$ introduce dimensionless bulk variables and a dimensionless slope,

$$A = \frac{a_2}{a_1}, \quad G = \frac{a_3}{a_1}, \quad H = \frac{a_4}{a_1}, \quad \theta = e^{a_1 u}, \quad \frac{B}{a_1} = \begin{pmatrix} b_0 & c_0 \\ d_0 & -b_0 \end{pmatrix}, \quad (4.30)$$

and the two discriminants

$$\Sigma = A + H, \quad J = 4G - (H - A)^2. \quad (4.31)$$

The first measures the imbalance of the two off-diagonal amplitudes and the second is the discriminant of the conic that the admissible slopes lie on. The nine cells are those of Table 7.

On $a_1 \neq 0, \Sigma \neq 0$, the upper and mixed templates are

$$K_U^{(4)} = I_2 + \frac{c_0(\theta^2 - \theta^{-2})}{4} E_{12}, \quad b_0 = d_0 = 0, \quad (4.32)$$

and, with $4b_0 = (H - A)d_0$, $4c_0 = -Gd_0$ and $\mathcal{D}_m = d_0\{A\theta^4 - \Sigma\theta^2 + H\} - 8\theta^2$,

$$K_M^{(4)} = \begin{pmatrix} 1 & \frac{Gd_0(\theta^4 - 1)}{2\mathcal{D}_m} \\ -\frac{2d_0(\theta^4 - 1)}{\mathcal{D}_m} & \frac{d_0\{H\theta^4 - \Sigma\theta^2 + A\} - 8\theta^2}{\mathcal{D}_m} \end{pmatrix}. \quad (4.33)$$

Equation (4.32) is the boundary with no diagonal deformation at all; Eq. (4.33) is the one whose three slope coordinates are locked to a single free parameter by the two displayed relations.

When $J \neq 0$ the remaining slopes satisfy

$$d_0 = 0, \quad \mathcal{Q} := Gb_0^2 + (H - A)b_0c_0 + c_0^2 - 4(AH + G) = 0, \quad (4.34)$$

so the admissible first derivatives form a conic in the (b_0, c_0) plane rather than a line: this is the structural consequence of the two equal off-diagonal columns of Eq. (2.12). Away from the two points $(b_0, c_0) = (2, -2H)$ and $(-2, -2A)$, set

$$\begin{aligned} \mathcal{E} &= Hb_0^2\theta^4 + b_0c_0\theta^4 + 2Hb_0\theta^4 + 2c_0\theta^4 - \Sigma b_0^2\theta^2 + Ab_0^2 - b_0c_0 + 4\Sigma\theta^2 - 2Ab_0 + 2c_0, \\ \mathcal{N}_y &= -AHb_0^2\theta^4 + (H - A)b_0c_0\theta^4 + c_0^2\theta^4 + 2AHb_0^2\theta^2 + (A - H)b_0c_0\theta^2 + 2\Sigma c_0\theta^2 \\ &\quad - AHb_0^2 + (H - A)b_0c_0 + c_0^2, \\ \mathcal{N}_w &= -Ab_0^2\theta^4 + b_0c_0\theta^4 + 2Ab_0\theta^4 - 2c_0\theta^4 + \Sigma b_0^2\theta^2 - Hb_0^2 - b_0c_0 - 4\Sigma\theta^2 - 2Hb_0 - 2c_0, \end{aligned} \quad (4.35)$$

three polynomials in θ whose coefficients are the conic data; then

$$K_C^{(4)} = \begin{pmatrix} 1 & \frac{(\theta^4 - 1)\mathcal{N}_y}{2\theta^2\mathcal{E}} \\ 0 & -\frac{\mathcal{N}_w}{\mathcal{E}} \end{pmatrix}. \quad (4.36)$$

The two excluded points are not missing solutions but points at which this chart degenerates; they are covered by writing, with $\Theta = AH + G$,

$$\Pi_+ = 4\Theta\theta^4 - 2\Sigma^2\theta^2 + \Sigma^2, \quad \Pi_- = \Sigma^2\theta^4 - 2\Sigma^2\theta^2 + 4\Theta, \quad (4.37)$$

$$\begin{aligned} \Upsilon_+ &= 2\Sigma\Theta(\theta^4 + 1) - (5A^2H + 4AG + 2AH^2 + H^3)\theta^2, \\ \Upsilon_- &= 2\Sigma\Theta(\theta^4 + 1) - (A^3 + 2A^2H + 5AH^2 + 4GH)\theta^2, \end{aligned} \quad (4.38)$$

so that the two boundaries there are

$$K_{C,+}^{(4)} = \begin{pmatrix} 1 & -\frac{(\theta^4 - 1)\Upsilon_+}{2\theta^2\Pi_+} \\ 0 & \Pi_-/\Pi_+ \end{pmatrix}, \quad K_{C,-}^{(4)} = \begin{pmatrix} 1 & -\frac{(\theta^4 - 1)\Upsilon_-}{2\theta^2\Pi_-} \\ 0 & \Pi_+/\Pi_- \end{pmatrix}. \quad (4.39)$$

Their diagonal entries are reciprocal, which is the signature of the two vertical directions of the conic.

When $J = 0$ the conic degenerates into two lines, and exactly one of them is consistent with the actual first derivative:

$$d_0 = 0, \quad c_0 = -\frac{H - A}{2}b_0 + \Sigma. \quad (4.40)$$

Writing

$$\begin{aligned} \mathcal{D}_\ell &= b_0(\theta^2 - 1) + 2(\theta^2 + 1), & \mathcal{N}_\ell &= b_0(\theta^2 - 1) - 2(\theta^2 + 1), \\ \mathcal{F}_\ell &= \Sigma b_0^2(\theta^2 - 1)^2 - 4\Sigma(\theta^2 + 1)^2 + 8(H - A)b_0\theta^2, \end{aligned} \quad (4.41)$$

the boundary on that line is

$$K_\ell^{(4)} = \begin{pmatrix} 1 & -\frac{(\theta^4 - 1)\mathcal{F}_\ell}{4\theta^2\mathcal{D}_\ell^2} \\ 0 & \frac{\mathcal{N}_\ell^2}{\mathcal{D}_\ell^2} \end{pmatrix}. \quad (4.42)$$

The second algebraic line carries the wrong first derivative except at the already listed upper point, so it contributes no family; this is a case in which the derivative condition is necessary but not sufficient, and the check is the substitution back into [Eq. \(2.4\)](#).

On $\Sigma = 0$ every slope is admissible and one rational template covers them all:

$$K_{\text{ex}}^{(4)} = \begin{pmatrix} 1 & (\theta^4 - 1)\mathcal{Y}_e/\mathcal{D}_e \\ -2d_0(\theta^8 - 1)/\mathcal{D}_e & -\mathcal{W}_e/\mathcal{D}_e \end{pmatrix}, \quad (4.43)$$

with

$$\begin{aligned} \mathcal{D}_e &= Ad_0\theta^8 - 2Ad_0\theta^6 - 4b_0\theta^6 - 8\theta^6 + 2Ad_0\theta^2 + 4b_0\theta^2 - Ad_0 - 8\theta^2, \\ \mathcal{Y}_e &= A^2d_0\theta^4 + 2Ab_0\theta^4 - 2A^2d_0\theta^2 - 2c_0\theta^4 - 4Ab_0\theta^2 + A^2d_0 + 2Ab_0 - 2c_0, \\ \mathcal{W}_e &= Ad_0\theta^8 - 2Ad_0\theta^6 - 4b_0\theta^6 + 8\theta^6 + 2Ad_0\theta^2 + 4b_0\theta^2 - Ad_0 + 8\theta^2. \end{aligned} \quad (4.44)$$

Since $\mathcal{D}_e = -16$ at $\theta = 1$, the chart is regular at the origin, which is what makes [Eq. \(4.43\)](#) a legitimate normalized representative. On $H = -A$, $G = A^2$, $A \neq 0$ the stabilizer locus $d_0 = 0$,

TABLE 8. The complete Class-5 partition.

Condition	Complete family
$a \neq 0, (p, r) \neq (0, 0)$	K_5 , every slope
$a \neq 0, p = r = 0$	$K_5 = I_2 + uB$, every slope
$a = 0, p + r \neq 0$	K_5 , every slope
$a = 0, r = -p \neq 0$	twisted self-commuting families Eq. (4.51)
$a = p = r = 0$	every self-commuting family

$c_0 = Ab_0$ enlarges to

$$K(u) \sim I_2 + \psi(u) \begin{pmatrix} 0 & 1 \\ 0 & -2/A \end{pmatrix}, \quad \psi(0) = 0, \quad \psi'(0) = a_1 c_0, \quad (4.45)$$

and at $A = G = H = 0$ the corresponding enlargement is every $\text{diag}(1, \psi(u))$ with $\psi(0) = 1$ and $\psi'(0) = -2b$.

For the removable limit $a_1 = 0$, write $A = a_2$, $G = a_3$, $H = a_4$ and $\mathcal{D}_0 = (A + H)du^2 - 2bu - 2$. Away from the functional enlargements, every slope has the single rational boundary

$$K_0^{(4)}(u) = \begin{pmatrix} 1 & -\frac{2u(AHdu^2 + c)}{\mathcal{D}_0} \\ -\frac{2du}{\mathcal{D}_0} & \frac{(A + H)du^2 + 2bu - 2}{\mathcal{D}_0} \end{pmatrix}, \quad (4.46)$$

a ratio of quadratics rather than a polynomial, because the limit of the exponential chart keeps its denominator. If $A = H$ and $G \neq 0$, the locus $b = d = 0$ enlarges to $I_2 + \psi(u)E_{12}$ with $\psi(0) = 0$ and $\psi'(0) = c$. If $A = H$ and $G = 0$, put $\Gamma_4(u) = I_2 + AuE_{12}$; the complete family is

$$K(u) = \Gamma_4(-u) \{ \psi_1(u)I_2 + \psi_2(u)B_0 \} \Gamma_4(-u), \quad (4.47)$$

for a fixed nonscalar B_0 , together with the scalar branch, where $\psi_1(0) = 1$ and $\psi_2(0) = 0$; writing $B_0 = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & -B_{11} \end{pmatrix}$, its slope is $b = \psi_2'(0)B_{11}$, $c = \psi_2'(0)B_{12} - 2A$, $d = \psi_2'(0)B_{21}$. At $A = G = H = 0$ this is [Example 2.2](#) again. The single-formula branch [Eq. \(4.46\)](#) meets [Eq. \(4.47\)](#) exactly at

$$B_0 = \begin{pmatrix} b & c + 2A \\ d & -b \end{pmatrix}, \quad \psi_1(u) = \frac{1}{1 + bu - Adu^2}, \quad \psi_2(u) = u \psi_1(u), \quad (4.48)$$

so the two descriptions agree on their intersection and nothing is counted twice.

4.5. Class 5. Write $a = a_1$, $p = a_2$ and $r = a_3$. On the three cells where the rank is three, one cubic polynomial exhausts the boundaries:

$$K_5(u) = \begin{pmatrix} 1 + bu + \frac{p-r}{2}du^2 & cu - pr du^3 \\ du & 1 - bu + \frac{p-r}{2}du^2 \end{pmatrix}. \quad (4.49)$$

Every entry of [Eq. \(4.49\)](#) is determined by the slope and the bulk parameters, and every slope occurs, so the Class-5 boundary set off the degenerate loci is a three-parameter family of polynomial matrices. The complete partition is [Table 8](#).

On the fourth row the bulk matrix factorizes. With $\Gamma_5(u) = I_2 + puE_{12}$,

$$R_5|_{r=-p}(u) = (\Gamma_5(u) \otimes \Gamma_5(-u)) P, \quad (4.50)$$

TABLE 9. The complete Class-6 partition.

Condition	Complete family
$a \neq 0, Q \neq 0$	K_6 , every slope
$a \neq 0, Q = 0$	$K_6 = I_2 + uB$, every slope
$a = 0, Q \neq 0$	K_6 , every slope
$a = Q = 0$	every self-commuting family

which says that on this line the Class-5 bulk matrix is nothing but the exchange operator dressed by an entire twist on each leg. Substituting Eq. (4.50) into Eq. (2.4) and setting

$$K(u) = \Gamma_5(u)\Lambda(u)\Gamma_5(u) \quad (4.51)$$

turns the reflection equation into $[\Lambda(u), \Lambda(v)] = 0$, so Lemma 2.3 applies: modulo the scalar gauge $\Lambda = I_2 + \psi B_0$ for one fixed traceless direction. Before the quotient of Section 5 the direction ranges over the whole projective line of traceless matrices; after it, four representatives suffice,

$$E_{12} + E_{21}, \quad E_{21}, \quad \sigma_z, \quad E_{12}, \quad (4.52)$$

namely the two-sided, lower, semisimple and upper directions, together with the scalar branch Γ_5^2 . The polynomial family meets these at

$$\psi(u) = u, \quad B_0 = \begin{pmatrix} b & c - 2p \\ d & -b \end{pmatrix}, \quad (4.53)$$

and in general the slope of Eq. (4.51) is $B = 2pE_{12} + \psi'(0)B_0$: the twist itself contributes a fixed upper-triangular piece to the first derivative. At the origin of the parameter space $\Gamma_5 = I_2$ and the three types are scalar, semisimple $I_2 + \psi\sigma_z$, and nilpotent $I_2 + \psi E_{12}$.

4.6. Class 6. Write $a = a_1$ and $Q = a_2$. Off the point where the bulk matrix becomes the exchange operator, one quintic polynomial exhausts the boundaries:

$$K_6(u) = \begin{pmatrix} 1 + bu - 2aQdu^3 & cu - (4aQb + Q^2d)u^3 + 12a^2Q^2du^5 \\ du & 1 - bu + 2aQdu^3 \end{pmatrix}. \quad (4.54)$$

The odd powers are forced: the quadratic and quartic coefficients vanish identically, so the boundary is an odd deformation of the identity in the two diagonal entries and of cu in the upper one. The partition is Table 9.

On the last row the three types are again I_2 , $I_2 + \psi\sigma_z$ and $I_2 + \psi E_{12}$, with slope $B = \psi'(0)B_0$ for the chosen direction and zero on the scalar branch. The specialization of Eq. (4.54) at the origin is only the subfamily $\psi(u) = u$ and is not counted separately.

At $a = 0$ with $Q \neq 0$ the elimination can be run by hand and is worth displaying, because it shows how the polynomial form is forced rather than assumed. In the trace gauge, writing $p_0(u) = (k_1(u) - k_4(u))/2$, the system Eq. (3.3) gives successively

$$k_3(u) = du, \quad k_2(u) = -Q^2du^3 + 2Qbu^2 - 2Qp_0(u)u + cu, \quad 8Q(p_0(u) - bu) = 0. \quad (4.55)$$

The first equation fixes the lower entry, the second expresses the upper entry through the remaining unknown p_0 , and the third — available because $Q \neq 0$ — forces $p_0(u) = bu$, which closes the system and reproduces Eq. (4.54) at $a = 0$.

Proof of Theorem 4.1. Equation (3.3) is a necessary condition satisfied by every admissible boundary, derived in Section 3 from four unrestricted meromorphic entries. By Lemma 3.2 the exact radical, saturation and rate-collision computations partition each $\mathcal{A}_j \times \mathbb{C}^3$ into precisely the cells listed above, with no residual component. On every rank-three cell the displayed normalized matrix

spans the kernel of $\mathcal{C}_{R,B}$, so it is the only candidate in the scalar gauge; on every rank-two or rank-zero cell the specialized equations reduce, by the twists displayed there, to a commutator identity whose solution set [Lemma 2.3](#) exhausts. Sufficiency is separate from necessity, and is supplied by substituting each displayed family back into the two-variable identity [Eq. \(2.4\)](#), with independent symbols for the values of every arbitrary function at u and at v ; every residual vanishes identically. Finally [Section 3.3](#) promotes the equality of normalized germs at the origin to an identity of single-valued meromorphic functions on $\mathbb{C}$. The tables are therefore valid and exhaustive on all six full parameter varieties. $\square$

5. THE FIXED-BULK QUOTIENT

Theorem 5.1 (Orbits at fixed bulk data). *For each family of [Theorem 4.1](#), quotienting by [Eq. \(2.5\)](#) gives exactly the groups, orbit slices and stabilizers of this section. Every transformation used preserves the same literal bulk matrix projectively in the sense of [Eq. \(2.6\)](#); no parameter-changing similarity and no similarity that is singular on a degeneration appears. The stabilizer jumps, the overlaps between finite and arbitrary-function families, and the constant-bulk degenerations are all included.*

For Classes 1, 2, 4, 5 and 6, comparing determinants on both sides of [Eq. \(2.6\)](#) forces the bulk scalar φ to be identically one: the rescaling may not change the normalization of the bulk matrix at all. For Class 3 the generator equation determines the scalar uniquely instead, and the table below is for the literal source matrix.

Three families of similarity recur, and it is convenient to name them once:

$$U_\tau = \begin{pmatrix} 1 & \tau \\ 0 & 1 \end{pmatrix}, \quad S_{s,\tau} = \begin{pmatrix} 1 & \tau \\ 0 & s \end{pmatrix}, \quad D_s = \text{diag}(1, s). \quad (5.1)$$

They are the unipotent, Borel and torus subgroups of the upper-triangular matrices; which of them survives at a given bulk point is what the tables record. Their action on the slope is universal:

additive	$(b, c, d) \mapsto (b + \tau d, c - 2\tau b - \tau^2 d, d)$	(5.2)
affine	$(b, c, d) \mapsto (s(b + \tau d), c - 2\tau b - \tau^2 d, s^2 d)$	
torus	$(b, c, d) \mapsto (sb, c, s^2 d)$	
weighted Borel	$(b, c, d) \mapsto (s^2(b + \tau d), s(c - 2\tau b - \tau^2 d), s^3 d)$	

Each row of [Eq. \(5.2\)](#) is conjugation of the traceless first derivative by the corresponding group, combined with the spectral rescaling where one is allowed; the weights on b , c and d record how each entry scales. Writing $\Delta_B = b^2 + cd$ for the discriminant of the slope, which is minus its determinant and therefore a conjugation invariant, the orbit slices are:

- additive: $(0, \Delta_B/d, d)$ for $d \neq 0$, $(b, 0, 0)$ for $d = 0$, $b \neq 0$, and $(0, c, 0)$ for $b = d = 0$;
- affine: $(0, \Delta_B/d, 1)$ for $d \neq 0$, $(1, 0, 0)$ for $d = 0$, $b \neq 0$, and $(0, c, 0)$ for $b = d = 0$;
- torus: $(1, c, d/b^2)$ for $b \neq 0$, $(0, c, 1)$ for $b = 0$, $d \neq 0$, and $(0, c, 0)$ for $b = d = 0$;
- weighted Borel: $(0, \varsigma, 1)$ for $d \neq 0$ with $\varsigma^3 = \Delta_B^3/d^4$ modulo cube roots of unity; $(1, 0, 0)$ for $d = 0$, $b \neq 0$; $(0, 1, 0)$ for $b = d = 0$, $c \neq 0$; and the origin.

For an arbitrary-function family $\Lambda = I_2 + \psi B_0$ the group acts on the fixed direction B_0 and on ψ together, and the quotient is a list of projective direction types each carrying a residual functional identity, as in [Tables 10](#) and [11](#).

At a bulk point where the matrix is the exchange operator, the only nonscalar directions are semisimple and nilpotent, with residual relations $\psi(u) \sim \pm \psi(\lambda u)$ and $\psi(u) \sim \varsigma \psi(\lambda u)$ respectively, the two constants in the second being independent. In every row of [Tables 10](#) and [11](#) the stabilizer of a family is exactly the subgroup for which the displayed relation holds, and the scalar branch is fixed by the whole group.

TABLE 10. Quotient of an arbitrary-function fiber by the affine or weighted-Borel group. The residual relation is the identification that survives on the free function.

Direction cell	Representative	Residual relation
$d \neq 0, \Delta_B \neq 0$	$E_{12} + E_{21}$	$\psi(u) \sim -\psi(-u)$
$d \neq 0, \Delta_B = 0$	E_{21}	$\psi(u) \sim s\psi(su)$
$d = 0, b \neq 0$	σ_z	$\psi(u) \sim \psi(su)$
$b = d = 0, c \neq 0$	E_{12}	$\psi(u) \sim s^{-1}\psi(su)$

TABLE 11. Quotient of an arbitrary-function fiber by a torus normalizer, written in the eigenbasis where $\tilde{B}_0 = n_0\sigma_z + e_+E_{12} + e_-E_{21}$.

Direction cell	Representative	Residual relation
$n_0 \neq 0, e_+e_- \neq 0$	$\sigma_z + E_{12} + \iota E_{21}, \iota = e_+e_-/n_0^2$	$\psi(u) \sim -\psi(-u)$
$n_0 \neq 0$, exactly one of $e_{\pm}$ nonzero	$\sigma_z + E_{12}$	none
$n_0 \neq 0, e_+ = e_- = 0$	σ_z	$\psi(u) \sim -\psi(-u)$
$n_0 = 0, e_+e_- \neq 0$	$E_{12} + E_{21}$	$\psi(u) \sim \pm\psi(\pm u)$
$n_0 = 0$, exactly one of $e_{\pm}$ nonzero	E_{12}	$\psi(u) \sim \varsigma\psi(u)$

TABLE 12. Class-1 fixed-bulk automorphisms, as pairs $(\lambda, [S])$.

Bulk condition	Complete group
$a_5 \neq 0$, on Z_1	$(1, [N_1 D_s N_1^{-1}])$ and $(-1, [N_1 W D_s N_1^{-1}])$
$a_5 \neq 0$, off Z_1	trivial
$a_5 = 0$, all remaining parameters zero	every λ and all of PGL_2
$a_5 = 0$, some parameter nonzero, $\ell_1 = 0$	$(s, [S_{s,\tau}])$
$a_5 = 0, \ell_1 \neq 0$	$(s, [D_s])$

5.1. **Classes 1 and 2.** For Class 1 set $Z_1 = \{a_4 = a_1 = t, a_2 = a_3 = -t\}$ and $\ell_1 = a_3 + a_4 - a_1 - a_2$; on Z_1 put $N_1 = \begin{pmatrix} 1 & t/a_5 \\ 0 & 1 \end{pmatrix}$ and write $W = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ for the swap. The complete group table is Table 12.

For Class 2, on $a_3 = \varepsilon = 0$ with $a_1 \neq 0$ put

$$N_2 = \begin{pmatrix} 1 & a_2/a_1 \\ 0 & 1 \end{pmatrix}, \quad W_2 = \begin{pmatrix} 1 & -2a_2/a_1 \\ 0 & -1 \end{pmatrix}, \quad (5.3)$$

the conjugator that straightens the bulk matrix and the involution it produces. The group table is Table 13.

The routing is as follows. On the first Class-1 row use Table 11 with $\tilde{B}_0 = N_1^{-1}B_0N_1$; the second row has trivial group, so every raw family there is already a normal form; the third uses the constant-bulk rules; the fourth uses Table 10 on $\mathcal{E}_1$ and the affine slices off it; the fifth uses the torus slices, with two exceptional rows,

$$K = I_2 + \psi(u)E_{12}, \quad \psi(u) \sim s^{-1}\psi(su), \quad a_1 = a_4 = 0, (a_2, a_3) \neq (0, 0), a_2 \neq a_3, \quad (5.4)$$

$$K = \text{diag}(1, \psi(u)), \quad \psi(u) \sim \psi(su), \quad a_2 = a_3 = 0, (a_1, a_4) \neq (0, 0), a_1 \neq a_4, \quad (5.5)$$

whose stabilizers are exactly the sets of s satisfying the displayed functional identity, the constant members $\psi = 0$ and $\psi = 1$ being fixed by the whole torus.

TABLE 13. Class-2 fixed-bulk automorphisms, as pairs $(\lambda, [S])$.

Bulk condition	Complete group
$a_1 \neq 0, a_3 = \varepsilon = a_5 = 0$	two N_2 -conjugated normalizer components, with $\lambda = \pm 1$
$a_1 \neq 0, a_3 = \varepsilon = 0, a_5 \neq 0$	$(1, [I_2])$ and $(1, [W_2])$
$a_1 \neq 0$, all other points	trivial
$a_1 = 0$, all remaining parameters zero	every λ and all of PGL_2
$a_1 = 0, a_2 = a_3 = a_4 = 0, a_5 \neq 0$	$(s^2, [S_{s,\tau}])$
$a_1 = 0$, nonzero linear part, $a_5 = 0$	$(s, [S_{s,\tau}])$
$a_1 = 0$, nonzero linear part, $a_5 \neq 0$	$(1, [U_\tau])$

TABLE 14. Class-3 fixed-bulk automorphisms.

Bulk stratum	Complete set of pairs $(\lambda, [S])$
$m = n = 0$, any g	every $\lambda \neq 0$ and every $[S] \in PGL_2$
$g \neq 0, (m, n) \neq (0, 0)$	trivial
$g = 0, (m, n) \neq (0, 0), m \neq \pm n$	$\lambda = 1, S = D_s$
$g = 0, m = n \neq 0$	$\lambda = 1, S$ diagonal or antidiagonal
$g = 0, m = -n \neq 0$	$\lambda = 1$ with S diagonal, or $\lambda = -1$ with S antidiagonal

For Class 2, on the first row use Table 11 with $\tilde{B}_0 = N_2^{-1}B_0N_2$. On the second row put $\hat{c} = c + 2a_2b/a_1$; the involution fixes b and sends $\hat{c} \mapsto -\hat{c}$, so the unique rows are labelled by $(b, \hat{c}^2)$, with stabilizer of order two exactly at $\hat{c} = 0$. At $b = -a_1$, if ψ is the arbitrary upper entry, set

$$\tilde{\psi}(u) = \psi(u) - \frac{a_2}{a_1} (e^{a_1 u} - e^{-a_1 u}); \quad (5.6)$$

subtracting this fixed particular solution makes the residual relation homogeneous, $\tilde{\psi} \sim -\tilde{\psi}$, whose stabilizer has order two exactly when $\tilde{\psi} = 0$. The remaining rows use the constant-bulk rules, the weighted-Borel slices, the Borel table on $\varepsilon = 0$, the affine slices on $a_3 = 0, \varepsilon \neq 0$, and the additive slices otherwise; at $b = d = 0$ with $a_2 = a_3 = a_4 = 0, a_5 \neq 0$ the orbit is

$$K = I_2 + \psi(u)E_{12}, \quad \psi(u) \sim s^{-1}\psi(s^2u), \quad (5.7)$$

and every τ acts trivially there.

5.2. Class 3. With (m, n, g) as in Eq. (4.15), the fixed-bulk groups are those of Table 14: a nonzero coupling g rigidifies the bulk matrix completely, and only at $g = 0$ does a torus survive.

On the diagonal-torus cell the slope transforms as $(b, c, d) \mapsto (b, c/s, sd)$, so the two-sided directions have slice $(b, 1, cd)$ while the upper and lower one-sided directions have $(b, 1, 0)$ and $(b, 0, 1)$. At $m = n$ the extra antidiagonal element sends $(b, c, d) \mapsto (-b, d/s, sc)$, identifying b with $-b$ on the two-sided slice and sending a diagonal family to its reciprocal.

On $g = 0, m = -n \neq 0$, the homogeneous family Eq. (4.29) has two projective direction cells. For $\gamma_1\gamma_2 \neq 0$ take $\gamma_1 = \gamma_2 = 1$; the residual equivalence is generated by

$$(\psi, \nu) \mapsto (-\psi, -\nu), \quad (\psi(u), \nu) \mapsto \left(\frac{\theta(u)^2\psi(-u)}{1 + \nu\theta(u)\psi(-u)}, -\nu \right), \quad (5.8)$$

the second of which is the reflection $u \mapsto -u$ acting through a Möbius transformation on the free function. For $\gamma_1\gamma_2 = 0$ the upper and lower charts merge; taking $\gamma_1 = 1, \gamma_2 = 0$ and using $(\psi, \nu) \mapsto (\psi/s, s\nu)$ leaves the two slices $\nu = 0$ and $\nu = 1$. A diagonal family obeys $\psi(u) \sim \psi(-u)^{-1}$.

TABLE 15. Class-4 fixed-bulk automorphisms.

Bulk condition	Complete group
$a_1 \neq 0, a_2 + a_4 \neq 0$	trivial
$a_1 \neq 0, a_2 + a_4 = 0, a_1 a_3 + a_2 a_4 \neq 0$	the identity and $\begin{pmatrix} 1 & a_2/a_1 \\ 0 & -1 \end{pmatrix}$, with $\lambda = 1$
$a_1 \neq 0, a_2 + a_4 = 0, a_1 a_3 + a_2 a_4 = 0$	a conjugated diagonal-torus normalizer, $\lambda = 1$
$a_1 = 0, a_2 \neq a_4$	a conjugated torus, spectral scale $\lambda = s$
$a_1 = 0, a_2 = a_4 \neq 0, a_3 \neq 0$	the additive group, $\lambda = 1$
$a_1 = 0, a_2 = a_4 \neq 0, a_3 = 0$	the affine group, spectral scale $\lambda = s$
$a_1 = 0, a_2 = a_4 = 0, a_3 \neq 0$	the weighted Borel group, spectral scale $\lambda = s^2$
zero bulk	every λ and all of PGL_2

5.3. **Class 4.** The eight fixed-bulk strata are those of Table 15.

On the second row, the shifted slope coordinates

$$b_1 = b_0 + \frac{A}{2}d_0, \quad c_1 = c_0 - Ab_0 - \frac{A^2}{4}d_0, \quad d_1 = d_0 \quad (5.9)$$

diagonalize the involution, which acts as $(b_1, c_1, d_1) \mapsto (b_1, -c_1, -d_1)$; the shift is exactly the conjugation that removes the off-diagonal part of the automorphism. On the third row the torus acts by $(b_1, c_1, d_1) \mapsto (b_1, c_1/s, sd_1)$ and its Weyl coset exchanges c_1 and d_1 while sending $b_1 \mapsto -b_1$. On the fourth, after conjugation by $I_2 + \frac{a_3}{2(a_2 - a_4)}E_{12}$, the action is $(b_1, c_1, d_1) \mapsto (sb_1, c_1, s^2d_1)$. The remaining nonzero-bulk rows use the additive, affine and weighted-Borel actions of Eq. (5.2), and the zero-bulk row uses the three constant-bulk types.

5.4. **Classes 5 and 6.** The Class-5 group changes on four cells:

$$\begin{array}{l|l} a \neq 0, (p, r) \neq (0, 0) & (1, [U_\tau]) \\ a \neq 0, p = r = 0 & (1, PGL_2) \\ a = 0, (p, r) \neq (0, 0) & (s, [S_{s,\tau}]) \\ a = p = r = 0 & (\lambda, PGL_2). \end{array} \quad (5.10)$$

The pattern is that switching off a bulk parameter enlarges the group, from a single unipotent direction at a generic point to the whole of PGL_2 at the origin. On the polynomial cells the action is the additive or affine row of Eq. (5.2); on the cell where the bulk matrix is rational and the group is all of PGL_2 , the invariant is $\Delta_B = b^2 + cd$ and the orbits are the scalar, nonzero nilpotent and semisimple types. On the twisted cell the four directions Eq. (4.52) and their residual relations are those of Table 10, merging at the origin into the semisimple and nilpotent constant-bulk types.

For Class 6 the four groups are

$$\begin{array}{l|l} a \neq 0, Q \neq 0 & (1, [I_2]) \\ a \neq 0, Q = 0 & (1, PGL_2) \\ a = 0, Q \neq 0 & (s^{-1}, [D_s]) \\ a = Q = 0 & \text{every } \lambda \text{ and all of } PGL_2. \end{array} \quad (5.11)$$

On the torus cell the slope transforms as $(b, c, d) \mapsto (b/s, c, d/s^2)$, with slices $(1, c, d/b^2)$ for $b \neq 0$, $(0, c, 1)$ for $b = 0, d \neq 0$, and $(0, c, 0)$ for $b = d = 0$. The rational and constant fibers carry the same PGL_2 orbit types by Jordan type as above. These rules also state every degeneration: a group may enlarge as a bulk parameter vanishes, but the enlargement is never used to identify two different bulk points, which would be a change of model rather than a change of representative.

Proof of Theorem 5.1. Differentiating Eq. (2.6) at the origin gives an exact equation for the generator of the similarity in the tensor square, linear in the unknown entries of S and in λ . Comparing

determinants and spectra on the two sides restricts the spectral scale, and solving the generator equation on both projective charts of S , with the relevant denominators saturated, yields exactly the strata tabulated above; the exponential, hyperbolic or polynomial form of each bulk matrix makes the infinitesimal condition sufficient, and the one remaining Class-4 covariant is verified directly. Conjugating the traceless first derivative by the resulting groups gives Eq. (5.2) and the shifted Class-3 and Class-4 variants; solving the stabilizer equations gives the displayed slices. On an arbitrary-function fiber, undoing the twist reduces the action to that of the normalizer, Borel or PGL_2 on a single fixed traceless direction, whose orbit types are the semisimple, nilpotent, two-sided and one-sided shapes listed. Direct substitution confirms every inclusion and every overlap between a finite and an arbitrary-function family. No further identification at fixed bulk data remains. $\square$

6. CROSSING, LEFT BOUNDARIES, AND OPEN CHAINS

6.1. The crossing datum and what it is for. An open chain needs a boundary at each end. The standard way to obtain the second from the first is a crossing relation: constants $M \in GL_2(\mathbb{C})$ and $\rho \in \mathbb{C}$ and a nonzero meromorphic χ with

$$R_{12}^{t_1}(u) M_1 R_{21}^{t_1}(-u - 2\rho) M_1^{-1} = \chi(u) I_4. \quad (6.1)$$

Equation (6.1) is the algebraic form of particle–antiparticle exchange: it says the first-leg partial transpose of the bulk matrix is inverted, up to a scalar and a fixed conjugation, by the same matrix at a reflected and shifted rapidity. Given such a datum, a right boundary $\tilde{K}_R$ prescribes the left boundary

$$K_L(u) = \tilde{K}_R(-u - \rho)^t M. \quad (6.2)$$

For a chain of L sites, write the two monodromies and the double-row transfer matrix as

$$T_a(u) = R_{aL}(u) \cdots R_{a1}(u), \quad \hat{T}_a(u) = R_{1a}(u) \cdots R_{La}(u), \quad (6.3)$$

$$t_L(u) = \text{Tr}_a \left[K_{L,a}(u) T_a(u) K_{R,a}(u) \hat{T}_a(u) \right]. \quad (6.4)$$

The auxiliary line crosses all L sites, reflects at the right end through K_R , crosses back, and is closed at the left end by K_L under the trace over the auxiliary space; the commuting tower of the open chain is generated by t_L . We call the left map *usable* on a cell when $[t_L(u), t_L(v)] = 0$ for every $L \geq 2$ and every ordered pair $(\tilde{K}_R, K_R)$ of independently chosen classified right boundaries — not merely for one boundary, and not merely for short chains.

6.2. Classes 1–4 admit no crossing datum.

Proposition 6.1. *For every $j \in \{1, 2, 3, 4\}$ and every $\mathbf{a} \in \mathcal{A}_j$ there is no triple (M, ρ, χ) with M invertible and χ not identically zero satisfying Eq. (6.1).*

Proof. Up to an overall scalar, each of Eqs. (2.8) and (2.10) to (2.12) has the pattern

$$R(u) = \begin{pmatrix} 1 & \varrho_{12}(u) & \varrho_{13}(u) & \varrho_{14}(u) \\ 0 & 0 & \varrho_{23}(u) & \varrho_{24}(u) \\ 0 & \varrho_{32}(u) & 0 & \varrho_{34}(u) \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (6.5)$$

that is, a matrix whose first column and last row are those of the identity and whose middle block is antidiagonal. Its first-leg partial transpose is

$$R^{t_1}(u) = \begin{pmatrix} 1 & \varrho_{12}(u) & 0 & \varrho_{32}(u) \\ 0 & 0 & 0 & 0 \\ \varrho_{13}(u) & \varrho_{14}(u) & 0 & \varrho_{34}(u) \\ \varrho_{23}(u) & \varrho_{24}(u) & 0 & 1 \end{pmatrix}. \quad (6.6)$$

The second row and the third column of Eq. (6.6) vanish identically, so $R^{t_1}(u)$ has rank at most three at every rapidity and every bulk point. Suppose Eq. (6.1) held. Choose u away from the poles and zeros of the finitely many meromorphic functions involved and of χ . The left side is a product of $R^{t_1}(u)$ with invertible matrices, hence of rank at most three, while $\chi(u)I_4$ has rank four. The contradiction is uniform in the bulk parameters, so no crossing datum exists anywhere on those four varieties. $\square$

No left map, transfer matrix, or boundary Hamiltonian obligation of the form Eq. (6.2) is therefore triggered for Classes 1–4. This is a genuine structural statement about these models and not a failure of search: the partial transpose of Eq. (6.5) is singular by inspection.

6.3. Classes 5 and 6: crossing exists exactly at nonzero rate.

Proposition 6.2. *For Class 5 with $(a, p, r) = (a_1, a_2, a_3)$, a crossing datum exists if and only if $a \neq 0$, and is then unique up to a scalar multiple of M :*

$$M = \begin{pmatrix} 1 & 0 \\ (p-r)/(2a) & 1 \end{pmatrix}, \quad \rho = \frac{1}{2a}. \quad (6.7)$$

For Class 6 with $(a, Q) = (a_1, a_2)$, a crossing datum exists if and only if $a \neq 0$, uniquely with

$$M = I_2, \quad \rho = \frac{1}{2a}. \quad (6.8)$$

For both source-normalized matrices the crossing scalar is

$$\chi(u) = 4au(au-1)(au+1)(au+2). \quad (6.9)$$

The shift $\rho = 1/(2a)$ is the pole of the bulk prefactor: crossing reflects the rapidity about the point where the bulk matrix degenerates. Equation (6.9) shows that the crossing scalar vanishes at the origin and at three further points, all of them fixed by the rate. Existence and uniqueness follow from exact elimination of the coefficients of Eq. (6.1) under the inequation $\det M \neq 0$; at $a = 0$ the elimination has no solution, and on the remaining constant and twisted fibers the rank of R^{t_1} supplies the same obstruction as in Proposition 6.1.

Crossing existence is not enough for usability.

Proposition 6.3. *On Class 6 with $aQ \neq 0$ the left map Eq. (6.2) is not usable. Taking $K_R = I_2$ and the independent seed with slope $(b, c, d) = (1, 0, 0)$, and expanding $t_2(u) = \sum_{k \geq 0} t^{(k)} u^k$, one has*

$$[t^{(1)}, t^{(2)}]_{1,2} = 64Q^2 \neq 0. \quad (6.10)$$

On Class 5 with $a \neq 0$ and $p \neq r$ the ideal generated by the divided commutator coefficients, saturated by the inequation $a \neq 0$, is the unit ideal; and on $p = r \neq 0$ the independent upper seed gives

$$[t^{(1)}, t^{(2)}]_{0,2} = 16r^2 \neq 0. \quad (6.11)$$

Each display is a single matrix entry of a commutator that would have to vanish, computed for a chain of two sites and exhibited as a nonzero number. Together with Proposition 6.2 they leave exactly the rational cells

$$a \neq 0, \quad p = r = 0 \quad (\text{Class 5}), \quad a \neq 0, \quad Q = 0 \quad (\text{Class 6}). \quad (6.12)$$

6.4. All-length commutativity on the rational cells. On Eq. (6.12) both classes reduce to the same rational bulk matrix

$$R(u) = (1 - au)(P + 2au I_4), \quad (6.13)$$

the exchange operator plus a multiple of the identity, times a scalar; this is the isotropic point of both families. There every classified right boundary is $I_2 + uB_R$ with $\text{tr } B_R = 0$, and for an independently chosen seed with slope $\tilde{B}$ the prescribed left boundary is

$$K_L^{\text{can}}(u) = I_2 - (u + \rho)\tilde{B}^t. \quad (6.14)$$

The left boundary is thus the seed reflected about $-\rho$ and transposed, which is Eq. (6.2) written out for a linear boundary.

Theorem 6.4. *On the cells Eq. (6.12), for every $L \geq 2$ and every ordered pair of classified boundaries, $[t_L(u), t_L(v)] = 0$.*

Proof. For Eq. (6.13) the Yang–Baxter equation, regularity, the permutation and full-transpose symmetries, scalar unitarity, and the crossing identity Eq. (6.1) all hold, and the dual reflection equation holds for arbitrary and independent B_R and $\tilde{B}$ by direct substitution. The reflection-algebra argument of [18], which builds the commutator of two double rows by moving one train of bulk matrices through the other and uses the two reflection equations to reverse the order at the ends, then applies verbatim and gives the claim for every L . $\square$

Literal representatives, rather than canonically normalized ones, must be retained, because the scalar gauge is part of the classification. Writing $\tilde{K}_R(u) = \tilde{f}(u)(I_2 + u\tilde{B})$ and $K_R(u) = f_R(u)(I_2 + uB_R)$, linearity of Eq. (6.4) in each boundary gives

$$t_L(u) = \tilde{f}(-u - \rho) f_R(u) t_L^{\text{can}}(u). \quad (6.15)$$

A change of gauge therefore multiplies the whole transfer matrix by a scalar function and cannot affect commutativity; but it can move the expansion point. Exactly three cases occur at $u = 0$: if $\tilde{f}(-\rho)$ is finite and nonzero, $t_L(0)$ is invertible; if it vanishes, $t_L(0) = 0$; and if it has a pole, K_L and t_L have a pole at the origin of the same order. The admissible gauges $1 + 2au$ and $(1 + 2au)^{-1}$ realize the last two cases, and no logarithmic derivative is taken there.

6.5. The open-chain Hamiltonian and an exact normalization obstruction. For canonically normalized pairs on the rational cells, $t_L^{\text{can}}(0) = 2 I_{V \otimes L}$ and the two-site density is

$$h = PR'(0) = 2aP - aI_4 = a(\sigma_x \otimes \sigma_x + \sigma_y \otimes \sigma_y + \sigma_z \otimes \sigma_z), \quad (6.16)$$

the isotropic Heisenberg interaction with coupling a . Differentiating the permutation train at the origin gives

$$\mathcal{H}_L = t'_L(0) t_L(0)^{-1} = 2 \sum_{n=1}^{L-1} h_{n,n+1} + (B_R)_1 - (\tilde{B}^t)_L, \quad (6.17)$$

modulo a multiple of the identity: the open chain is the bulk sum with one one-site term at each end, the right end carrying the slope of the right boundary and the left end the transposed slope of the seed. Finite nonzero literal gauges change only the additive scalar.

The bulk sum in Eq. (6.17) carries a coefficient two rather than one, and this cannot be repaired by adjusting the endpoints.

Proposition 6.5. *There is no choice of one-site operators at sites 1 and L , and no multiple of the identity, for which Eq. (6.17) equals $\sum_n h_{n,n+1}$ plus those one-site terms.*

Proof. Take $L = 4$ and expand both sides in the basis of tensor products of Pauli matrices. The coefficient of $I_2 \otimes \sigma_x \otimes \sigma_x \otimes I_2$ in the excess h_{23} is $a \neq 0$, while an operator supported at site 1 or site 4 and a multiple of the identity both have coefficient zero there. $\square$

The factor two is the record of the two passages of the auxiliary line through every bulk bond in Eq. (6.4). It is a property of the unhalved logarithmic derivative used here, not a missing boundary term; halving it, as is customary, restores the conventional coefficient.

7. A CONSTANT LEFT BOUNDARY FOR EVERY CLASS-6 BOUNDARY

Proposition 6.3 says that no single crossing-induced left boundary works against every independently chosen right boundary off the rational cells. That is a stronger demand than an open chain makes. To build one commuting tower it is enough to exhibit, for each right boundary, one compatible left boundary. For Class 6 there is one, and it is the same for all of them.

Theorem 7.1. *Let R_6 be the source-normalized Class-6 matrix at any $(a, Q) \in \mathbb{C}^2$, let K_R be any admissible boundary classified in Section 4.6, including the arbitrary-function families at $a = Q = 0$, and let $K_L(u) = I_2$. Then for every $L \geq 2$ the double-row transfer matrix*

$$t_L(u) = \text{Tr}_a \left[T_a(u) K_{R,a}(u) \hat{T}_a(u) \right] \quad (7.1)$$

satisfies $[t_L(u), t_L(v)] = 0$ as an identity of meromorphic functions. If $K_R(0) = \kappa I_2$ with $\kappa \neq 0$, then $t_L(0) = 2\kappa I$ is invertible and

$$t'_L(0) t_L(0)^{-1} = 2 \sum_{n=1}^{L-1} h_{n,n+1} + (\kappa^{-1} K'_R(0))_1, \quad h = PR'_6(0). \quad (7.2)$$

Equation (7.2) says that the open Class-6 chain built this way is the bulk chain with a single one-site term at the boundary site, given by the derivative of the boundary matrix, and with no term at the auxiliary end at all. Note what the theorem does not say: it exhibits one compatible left boundary for each right boundary and does not assert that an arbitrary crossing-induced left boundary works at $aQ \neq 0$, which Proposition 6.3 refutes.

7.1. The Class-6 data. Cancel the removable denominator in the middle block of Eq. (2.14), keeping the source scalar; this gives

$$R_6(u) = (1 - au) \begin{pmatrix} 1 + 2au & Qu(1 + 2au) & Qu(1 + 2au) & -Q^2u^2(1 + 2au)^2 \\ 0 & 2au & 1 & -Qu(1 + 2au) \\ 0 & 1 & 2au & -Qu(1 + 2au) \\ 0 & 0 & 0 & 1 + 2au \end{pmatrix}, \quad (7.3)$$

a matrix polynomial in the rapidity of degree four. Two features are used below and both are visible in Eq. (7.3): the middle block is symmetric, so

$$R_{21}(u) = R_{12}(u), \quad (7.4)$$

meaning that Class 6 is one of the families for which the two orderings of Eq. (2.4) agree; and braided unitarity holds in the sharp form

$$R_{12}(u) R_{21}(-u) = \varpi(u) I_4, \quad \varpi(u) = (au - 1)(au + 1)(2au - 1)(2au + 1), \quad (7.5)$$

so exchanging two excitations and exchanging them back is the identity up to an explicit quartic scalar. The determinants

$$\det R_6(u) = (au - 1)^4 (2au - 1)(2au + 1)^3, \quad \det R_6^{t_1}(u) = 16a^3u^3(au - 1)^4(au + 1) \quad (7.6)$$

show that R_6 and its first-leg partial transpose are invertible as meromorphic matrix functions exactly when $a \neq 0$; at $a = 0$ the partial transpose is identically singular, which is why that stratum needs a separate argument.

7.2. The generic stratum.

Proof of Theorem 7.1 for $a \neq 0$. Set $\tilde{R}(u) = ((R(u)^{t_1})^{-1})^{t_1}$, which by Eq. (7.6) is a well-defined meromorphic matrix function for $a \neq 0$. The boundary transfer-matrix theorem of [29, Thm. 2.4] derives commutativity of two boundary transfer matrices under three hypotheses: that R and R^{t_1} are invertible as meromorphic matrix functions, which Eq. (7.6) supplies for $a \neq 0$; the right boundary-monodromy reflection equation; and an operator-valued dual reflection equation for the left boundary. For a one-dimensional left space and $K_L = I_2$ the latter reduces to

$$R_{21}(u-v)^{-t} \tilde{R}_{12}(u+v)^t = \tilde{R}_{21}(u+v)^t R_{12}(u-v)^{-t}, \quad (7.7)$$

where X^{-t} abbreviates $(X^{-1})^t$. Direct substitution of Eq. (7.3) into Eq. (7.7) gives a zero 4×4 residual in the unrestricted symbols a , Q , $u-v$ and $u+v$: the constant left boundary satisfies the dual equation identically, with no crossing relation, no PT hypothesis, and no restriction on Q entering.

It remains to connect the theorem's dressed monodromy to Eq. (7.1). The permutation symmetry Eq. (7.4) and the local Yang–Baxter equation give the global Yang–Baxter relation for $\hat{T}_a$, and braided unitarity Eq. (7.5) gives, meromorphically,

$$\hat{T}_a(-u)^{-1} = \varpi(u)^{-L} T_a(u), \quad (7.8)$$

that is, reversing the auxiliary line and inverting it returns the other monodromy up to the L -th power of the unitarity scalar. Dressing the right boundary in the standard way therefore produces

$$\mathcal{K}_a(u) = \hat{T}_a(-u)^{-1} K_{R,a}(u) \hat{T}_a(u) = \varpi(u)^{-L} T_a(u) K_{R,a}(u) \hat{T}_a(u), \quad (7.9)$$

which obeys the right boundary-monodromy reflection equation because K_R obeys Eq. (2.4). Applying the theorem to Eqs. (7.7) and (7.9) gives a commuting family $\text{Tr}_a \mathcal{K}_a(u) = \varpi(u)^{-L} t_L(u)$, and multiplying by the scalar meromorphic function $\varpi(u)^L$ leaves the commutator unchanged. Hence $[t_L(u), t_L(v)] = 0$ for every L . Multiplying K_R by an admissible scalar gauge multiplies t_L by the same function, so the conclusion holds for every literal representative and not only for the trace-normalized formula. $\square$

Remark 7.2. The scalar specialization of the dual reflection equation as displayed in [29, Eq. (2.13)] ends in R_{21}^{-t} , whereas transposing the operator equation of [29, Thm. 2.4] in full ends in R_{21}^{-1} , which is the form used in that paper's own appendix. Substituting Eq. (7.3) separates the two: the operator equation and the form ending in R_{21}^{-1} both give a zero residual, while the printed specialization leaves fifteen nonzero entries for generic Q . The proof above uses the operator equation directly and so assumes nothing about the printed specialization.

7.3. The singular strata.

Proof of Theorem 7.1 at $a = 0$. Fix L and take first $Q \neq 0$. For the classified boundary Eq. (4.54), every entry of the commutator

$$C_L(a, Q, b, c, d; u, v) = [t_L(u), t_L(v)] \quad (7.10)$$

is a polynomial in all seven displayed variables, because Eqs. (4.54) and (7.3) are polynomial and Eq. (7.1) is a finite product and trace. The previous subsection proves that this polynomial vanishes for every $a \neq 0$ with Q, b, c, d, u, v unrestricted. The complement of $a = 0$ is Zariski dense in $\mathbb{C}$, so every entry of C_L is the zero polynomial, and evaluating at $a = 0$ proves commutativity on that stratum. Since L was arbitrary, this is an all-length continuation and not an inference from finite-length checks.

At $a = Q = 0$ the bulk matrix is P and Eq. (2.4) reads $[K_R(u), K_R(v)] = 0$, so the admissible boundaries are the families of Lemma 2.3. There $\hat{T}_a(0) = T_a(0)^{-1}$, and conjugating by the

permutation train moves the auxiliary boundary onto the first site,

$$T_a(u)K_{R,a}(u)\widehat{T}_a(u) = (K_R(u))_1, \quad (7.11)$$

after which the auxiliary factor is idle and the trace contributes a factor two:

$$t_L(u) = 2(K_R(u))_1. \quad (7.12)$$

The transfer matrix is thus the boundary matrix itself, acting on one site. Its commutator is $4[K_R(u), K_R(v)]_1 = 0$, which covers the scalar, semisimple and nilpotent arbitrary-function families alike, including those whose first nonzero term occurs at arbitrarily high order. $\square$

7.4. The logarithmic derivative.

Proof of Eq. (7.2). For Eq. (7.3) the two-site density is

$$h = PR'_6(0) = \begin{pmatrix} a & Q & Q & 0 \\ 0 & -a & 2a & -Q \\ 0 & 2a & -a & -Q \\ 0 & 0 & 0 & a \end{pmatrix} = a(\sigma_x \otimes \sigma_x + \sigma_y \otimes \sigma_y + \sigma_z \otimes \sigma_z) + Q(\sigma_z \otimes E_{12} + E_{12} \otimes \sigma_z), \quad (7.13)$$

an isotropic exchange of strength a plus a non-Hermitian term of strength Q that raises one spin while measuring the other. Both partial traces of h vanish, which is the fact used at the end of this proof.

At $u = 0$ the two permutation monodromies cancel and $K_R(0) = \kappa I_2$, so $t_L(0) = 2\kappa I$. Differentiating Eq. (7.1) at the origin before taking the auxiliary trace separates the terms into three groups. Differentiating the boundary conjugates $K'_R(0)$ from the auxiliary factor onto site 1 and, after the trace, contributes $2(K'_R(0))_1$. For each internal bond $(n, n+1)$, differentiating the forward and the reflected copy of R gives two equal contributions, together $4\kappa h_{n,n+1}$. The two remaining terms, in which the differentiated bulk matrix sits at an end of the train adjacent to the auxiliary space, reduce after the trace to partial traces of h and vanish. Hence

$$t'_L(0) = 4\kappa \sum_{n=1}^{L-1} h_{n,n+1} + 2(K'_R(0))_1, \quad (7.14)$$

and dividing by $t_L(0) = 2\kappa I$ gives Eq. (7.2). $\square$

As on the rational cells, the bulk sum carries a coefficient two, for the same reason: the auxiliary line traverses every bond twice. Halving the logarithmic derivative gives the conventional normalization, with the boundary term halved along with it.

No analogous construction is claimed for Classes 1–5. The argument above uses three specific facts about Class 6 — the invertibility of the partial transpose at nonzero rate, the polynomiality that permits continuation in the rate, and the vanishing of both partial traces of the density — and each has to be re-examined before the same route is tried elsewhere.

8. INDEPENDENT CHECKS

The classification rests on the exact eliminations of Lemma 3.2 and Theorem 4.1 and on the continuation argument of Section 3.3. Alongside these, several independent computations were run whose purpose is to catch an error rather than to establish a theorem, and it is worth being explicit about which is which.

Every family printed in Section 4 was substituted back into the two-variable identity Eq. (2.4) in exact arithmetic, with the bulk scalars restored and with independent symbols for the values of every arbitrary function at u and at v . This establishes sufficiency, which the differentiated system does not; it would not by itself establish exhaustiveness. Across the six classes the substitution covers 26 cells and 11 boundary templates for Classes 1 and 2, 10 cells and 19 families for Class 3

TABLE 16. Independent power-series regressions through order twelve. Ranks are those of the extension system at each order, constant over the eleven orders tested.

Class	Representatives	Extension steps	Rank 0	Rank 2	Rank 3
1	124	1364	8	71	45
2	124	1364	10	58	56
3	52	572	2	18	32
4	67	737	6	26	35
5	17	187	2	6	9
6	10	110	1	2	7
Total	394	4334	29	181	184

with all 23 cell–family incidences, 9 cells with 13 families and 14 overlaps for Class 4, and the five and four cells of Classes 5 and 6 with their orbit representatives; every residual is identically zero.

Second, a power-series regression was run independently of the elimination. In the gauge $k_1(u) = 1$, all three remaining matrix coefficients at each order $n = 2, \dots, 12$ were introduced as unrestricted variables, the reflection equation was expanded through total degree $n + 1$, the classified extension was checked against the general solution, and the completed series was then substituted directly through total degree 13. The totals are those of Table 16: 394 representative bulk and slope points, 4334 extension steps, every step exact and every completed residual zero. A rank of three at an order means the next coefficient is unique in the fixed gauge, a rank of two means one coefficient remains free, and a rank of zero marks the points at which a new commuting direction can begin at higher order. These are diagnostics. They cannot exclude a branch that first separates above order twelve, and they say nothing about global continuation; the exact differentiated-system, functional-equation and meromorphic-identity arguments carry those steps.

Third, the double-row transfer matrices of Eqs. (6.4) and (7.1) were built explicitly and their commutators computed at $L = 2, 3, 4$ and 5, both on the rational cells of Eq. (6.12) and, for Class 6, at a generic point $(a, Q) = (2, 3)$ and at the singular point $(0, 3)$. All vanish. These finite-length checks are regressions on the all-length arguments of Theorems 6.4 and 7.1, not substitutes for them; conversely the two nonzero entries of Eqs. (6.10) and (6.11) are counterexamples, and a counterexample at one length is conclusive in a way a confirmation at one length is not.

Finally, the objects themselves were reconstructed independently of the classification: the six bulk matrices and their removable limits, the Yang–Baxter equation, regularity, braided unitarity, the crossing identity Eq. (6.1), the dual identity Eq. (7.7), the density Eq. (7.13) and its two partial traces were each verified symbolically from the definitions.

9. DISCUSSION AND LIMITATIONS

The classification splits the six families along a line that was not visible from the bulk. Classes 1–4 share the pattern Eq. (6.5), and that pattern alone destroys crossing: their partial transposes are singular, so no crossing datum exists at any parameter value, and the standard route to a left boundary is closed for them entirely. Classes 5 and 6 are rational rather than exponential, possess crossing wherever their rate is nonzero, and there the obstruction is of a different kind — not the absence of the datum but the failure of the left boundary it prescribes to work against every independently chosen right boundary. That failure is exact and quantitative, Eqs. (6.10) and (6.11), and it disappears precisely at the isotropic points Eq. (6.12), where the bulk matrix is the exchange operator plus a multiple of the identity.

What [Theorem 7.1](#) then shows is that the failure was an artefact of asking for too much. An open chain needs one left boundary per right boundary, not a map that works for every pair, and for Class 6 the constant matrix supplies it at every bulk point. The three ingredients are the invertibility of the partial transpose at nonzero rate, which brings the dual reflection equation into play; polynomiality in the rate, which carries the conclusion onto the stratum where that invertibility fails; and the vanishing of both partial traces of the density, which is what removes the auxiliary-end term from [Eq. \(7.2\)](#). Where those three hold elsewhere, the same route should work; the natural next case is Class 5, whose bulk matrix is the nearest structural neighbour, and it is not settled here.

Seven limitations bound what has been established.

- (1) The load-bearing eliminations are computer assisted. Their specification is in the text — the objects, the test, and the coverage argument of [Lemma 3.2](#) — so the proof can be followed without running anything, but the ideals themselves were computed rather than derived by hand.
- (2) The computations have been checked internally and re-derived from the definitions by independent programs, but no external group has reproduced them.
- (3) The literature audit behind the novelty statement is broad and dated, and cannot exclude an unindexed or unpublished treatment of the same six families.
- (4) The power-series regressions cannot exclude a branch that first separates above order twelve, and they do not address global continuation. Those steps rest on the arguments of [Sections 3](#) and [3.3](#).
- (5) On the generic Class-5 and Class-6 cells, crossing data exist but the prescribed left map is not usable. This does not classify alternative dual constructions outside the crossing convention [Eq. \(6.1\)](#), and [Theorem 7.1](#) is one such alternative.
- (6) On the rational cells, some admissible literal gauges make $t_L(0)$ zero or singular, as [Eq. \(6.15\)](#) shows; the logarithmic derivative is asserted only in the finite nonzero case.
- (7) The coefficient two in [Eqs. \(6.17\)](#) and [\(7.2\)](#) belongs to the unhalved logarithmic derivative adopted here. It is exact, and [Proposition 6.5](#) shows that no endpoint operator removes it; the customary half-normalization does.

Two questions are left open and both are concrete. Does the fixed left boundary of [Theorem 7.1](#) have an analogue for Classes 1–5, and if not, what replaces it at a wall for the four classes that have no crossing datum at all? And what do the resulting open chains look like spectrally: Classes 5 and 6 have non-diagonalizable densities, [Eq. \(7.13\)](#) being nilpotent in its Q part, so the boundary terms of [Eq. \(7.2\)](#) act on a Jordan structure rather than on a spectrum, and the effect of a boundary on that structure is not known. There is precedent to draw on. The generalized eigenvectors of a non-diagonalizable chain can be recovered as the limit of the eigenvectors of a diagonalizable perturbation of it, as has been carried out for the eclectic chain [\[30\]](#); and the quantum-group-invariant open XXZ chain at a root of unity is a case where an open end and Jordan cells already coexist, and where the algebraic Bethe ansatz was extended to cover the generalized eigenvectors [\[31\]](#). Whether either construction survives a boundary of the kind classified here is open.

10. CONCLUSION

For each of the six regular difference-form 4×4 bulk R -matrices outside the eight-vertex list, and at every point of its full complex parameter variety, we have determined every regular single-valued meromorphic right reflection matrix, together with its orbit under the scalar gauge and the constant similarities and spectral rescalings that preserve the same literal bulk matrix. The derivation assumes no form for the boundary: one differentiation of the reflection equation at the origin makes the problem linear, and an exact stratification by the rank of the resulting system splits each parameter variety into finitely many cells carrying either a unique kernel or a centralizer

family of arbitrary functions. Every removable limit, singular stratum, arbitrary-function fiber and overlap is included, and every family is a single-valued meromorphic function on the plane.

The consequences for open chains are sharp and asymmetric. Four of the six classes admit no crossing datum at any parameter value, for the structural reason that their first-leg partial transposes drop rank; the other two admit a unique datum exactly at nonzero rate, and the left boundary it prescribes satisfies the every-pair requirement only at their isotropic points. Fixing the left boundary to the identity instead gives, for Class 6 and at every bulk point, a commuting double-row transfer matrix for every classified right boundary, whose logarithmic derivative is twice the bulk chain plus one one-site endpoint term.

What this opens is no longer a classification problem. The exact boundary families make the spectral, Jordan-block and continuum-limit analyses of these open chains concrete, and the six bulk models — several of them non-Hermitian and non-diagonalizable — can now be studied with a wall. The first thing to settle is whether the constant left boundary of [Theorem 7.1](#) has an analogue in the remaining five classes; a single Class-5 counterexample, at any bulk point and any chain length, would decide it in the negative.

ACKNOWLEDGMENTS

Z. Y. acknowledges support from the Strategic Priority Research Program of the Chinese Academy of Sciences under Grant No. XDB1680102. Z. Y. also acknowledges support by National Natural Science Foundation of China (Grant No. 12247101). We also acknowledge Gewu Intelligence Lab for providing the collaborative research environment in which this project was developed.

OpenAI GPT-5.6 Sol was used substantively in this work: in solving the problem, including the formulation of arguments and the derivation of intermediate steps, and in drafting and revising the manuscript. The authors specified the problem, guided the approach to be taken, checked every argument and result it produced, and take full responsibility for the content of this manuscript.

CODE AVAILABILITY

Four steps of the argument are computer assisted: the constructible stratification of [Lemma 3.2](#), on which [Theorem 4.1](#) rests; the tensor-square generator computation behind [Theorem 5.1](#); the coefficient elimination establishing [Proposition 6.2](#); and the substitution verifying that the constant left boundary satisfies the dual reflection equation [Eq. \(7.7\)](#), which is the hypothesis discharged in the proof of [Theorem 7.1](#). Each is specified in the text — the objects, the test applied to them, and why the cases are exhaustive — so that the proof can be followed without running anything. The exact symbolic programs that carry out these computations, and the programs performing the independent checks of [Section 8](#), are available from the author on request.

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